Chapter 9: Random Processes – Part II

9.6 Stationary Random Processes

9.61 a)

$$\begin{aligned} X(t) &= A\cos 2\pi t \\ m_X(t) &= \mathcal{E}[A\cos 2\pi t = 0 \\ C_X(t_1,t_2) &= VAR[A]\cos 2\pi t_1 \cos 2\pi t_2 \quad \text{from Example 9.9} \\ &= \frac{1}{3}\cos 2\pi t_1 \cos 2\pi t_2 \end{aligned}$$

Autocovariance does not depend only on $t_1 - t_2$

$\Rightarrow X(t)$ not stationary, not wide sense stationary

b) $X(t) = \cos(\omega t + \Theta)$

From Example 9.10 $m_X(t) = 0$, $C_X(t_1,t_2) = \frac{1}{2}\cos\omega(t_1 - t_2)$

$\Rightarrow X(t)$ is wide sense stationary

In order to determine whether $X(t)$ is stationary, consider the third-order joint pdf:

$$\begin{aligned} &f_{X(t_1)X(t_2)X(t_3)}(x_1,x_2,x_3)dx_1dx_2dx_3 \\ &\quad = P[x_1 < \cos(\omega t_1 + \Theta) \le x_1 + dx_1,\ \ x_2 < \cos(\omega t_2 + \Theta) \le x_2 + dx_2, \\ &\qquad x_3 < \cos(\omega t_3 + \Theta) \le x_3 + dx_3] \\ &= P[A_1 \cap A_2 \cap A_3] \end{aligned}$$

where

$$A_i = \left\{ \cos^{-1} x_i - \omega t_i < \Theta \leq \cos^{-1} x_i - \omega t_i + \frac{dx_i}{\sqrt{1-x_i^2}} \right\}$$

see Example 4.36 and Figure 4.14.

$$\begin{aligned} &f_{X(t_1+\tau)X(t_2+\tau)X(t_3+\tau)}(x_1, x_2, x_3) \\ &\quad P[x_1 < \cos(\omega t_1 + \omega\tau + \Theta) \leq x_1 + dx_1, \\ &\quad\ x_2 < \cos(\omega t_2 + \omega\tau + \Theta) \leq x_2 + dx_2, \\ &\quad\ x_3 < \cos(\omega t_3 + \omega\tau + \Theta) \leq x_3 + dx_3] \\ &\quad = P[A_1' \cap A_2' \cap A_3'] \end{aligned}$$

where

$$A_i' = \left\{ \cos^{-1} x_i - \omega t_i - \omega\tau < \Theta \leq \cos^{-1} x_i - \omega t_i - \omega\tau + \frac{dx_i}{\sqrt{1-x_i^2}} \right\}$$

Since Θ is uniformly distributed, $P[A_i] = P[A_i']$.

In addition

$$P[A_1 \cap A_2 \cap A_3] = P[A_1' \cap A_2' \cap A_3']$$

since the intersection depends only on the relative values of t_1, t_2 and t_3. The same procedure can be used for nth order pdf's.

$\therefore X(t)$ is a stationary random process.

9.62

Head : X_n: ... 1 1 1 ...

Tail: X_n: ... -1 -1 -1 ...

a) $E\{X_n\} = P\{\text{Head}\} \times 1 + (-1) \times P\{\text{Tail}\} = 0$

$$C_X(n_1, n_2) = E[X_{n_1} X_{n_2}] - E[X_{n_1}]E[X_{n_2}] = E[X_{n_1} X_{n_2}]$$

$$= 1 \times 1 \times P\{\text{Head}\} + (-1) \times (-1) \times P\{\text{Tail}\} = \frac{1}{2} + \frac{1}{2} = 1$$

Therefore X_n is WSS

b) $VAR\{X_n\} = E[X_n^2] - E[X_n]^2 = 1 - 0 = 1$

$R_X(n_1, n_2) = 1$

for third order pmf we have

$$P\{X_{n_1} = k_1, X_{n_2} = k_2, X_{n_3} = k_3\} = \begin{cases} \frac{1}{2} & k_1 = k_2 = k_3 = 1 \\ \frac{1}{2} & k_1 = k_2 = k_3 = -\frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

it is not dependent on sample difference and placement of origin

True to the nth order,

then X_n is stationary.

c) X_n stationary $\Rightarrow$ X_n cyclostationary with period $\tau = 1$

9.63)

a) Head: X_n : .... -1 1 -1 1 ... $(-1)^n$

Tail: X_n : .... 1 -1 1 -1 .. $(-1)^{n+1}$

$$E[X_n] = (-1)^n \times \frac{1}{2} + (-1)^{n+1}\left(\frac{1}{2}\right) = 0$$

$$R_X(n_1,n_2) = E[X_{n_1}X_{n_2}] = \begin{cases} -1 & \text{if } n_2-n_1 \text{ odd} \\ 1 & \text{if } n_2-n_1 \text{ even} \end{cases}$$

$$C_X(n_1,n_2) = R_X(n_1,n_2)$$

Therefore:
X_n is WSS

b) $VAR[X_n] = E[X_n^2] - E^2[X_n] = 1 \times \frac{1}{2} + 1 \times \frac{1}{2} - 0 \times 0 = 1$

we have to show that:

$$P\{X_{n_1}=k_1, X_{n_2}=k_2, X_{n_3}=k_3\} = P\{X_{n_1+m}=k_1, X_{n_2+m}=k_2, X_{n_3+m}=k_3\}$$

if m is even the above equality holds trivially.

if m is odd the above equality is also true.

This can be shown for nth order pmf this is true as well.

Then X_n is stationary.

c) X_n is cyclostationary with period $T=1$.

9.64

$$f_{X(t_1)X(t_2)\ldots X(t_n)}(x_1, x_2, \ldots, x_k)$$
$$= f_{X(t_1)\ldots X(t_k)}(x_2, \ldots, x_k|x_1) \underbrace{f_{X(t_1)}(x_1)}_{1 \text{ unif. dist.}}$$
$$= \prod_{i=2}^{k} \delta(g^{-1}(x_i) - (t_i - \tilde{t})) \cdot 1$$
$$\prod_{i=2}^{k} \delta(g^{-1}(x_i) - (t_i - t_1 + g^{-1}(x_1)))$$

where

$$\tilde{t} = t_1 - g^{-1}(x_1)$$

once x_1 is known the phase shift is determined unambiguously.

$$f_{X(t_1)\ldots X(t_k)}(x_1, \ldots, x_k) = \prod_{i=2}^{k} \delta(g^{-1}(x_i) - g^{-1}(x_1) - (t_2 - t_1))$$

$\therefore X(t)$ is a stationary random process.

9.65

~~6.53~~ $X(t) = A\cos\omega t + B\sin\omega t$

a)
$$\begin{aligned}
\mathcal{E}[X(t)] &= \mathcal{E}[A\cos\omega t + B\sin\omega t] \\
&= \mathcal{E}[A]\cos\omega t + \mathcal{E}[B]\sin\omega t = 0 \\
C_X(t_1,t_2) &= \mathcal{E}[(A\cos\omega t_1 + B\sin\omega t_1)(A\cos\omega t_2 + B\sin\omega t_2)] \\
&= \mathcal{E}[A^2]\cos\omega t_1\cos\omega t_2 + \mathcal{E}[B^2]\sin\omega t_1\sin\omega t_2 \\
&\quad +\mathcal{E}[A]\mathcal{E}[B]\cos\omega t_1\sin\omega t_2 + \mathcal{E}[A]\mathcal{E}[B]\sin\omega t_1\cos\omega t_2 \\
&= \mathcal{E}[A^2]\cos\omega t_1\cos\omega t_2 + \mathcal{E}[B^2]\sin\omega t_1\sin\omega t_2 \\
&= \mathcal{E}[A^2]\underbrace{(\cos\omega t_1\cos\omega t_2 + \sin\omega t_1\sin\omega t_2)}_{\frac{1}{2}\cos\omega(t_1-t_2)} \\
&\quad \text{where we assumed } \mathcal{E}[A^2] = \mathcal{E}[B^2] \\
&= \frac{1}{2}\mathcal{E}[A^2]\cos\omega(t_1-t_2)
\end{aligned}$$

$\therefore X(t)$ is WSS.

b)
$$\begin{aligned}
\mathcal{E}[X^3(t)] &= \mathcal{E}[(A\cos\omega t + B\sin\omega t)^3] \\
&= \mathcal{E}[A^3\cos^3\omega t + 3A^2B\cos^2\omega t\sin\omega t + 3AB^2\cos\omega t\sin^2\omega t \\
&\quad +B^3\sin^{-3}t] \\
&= \mathcal{E}[A^3]\cos^3\omega t + \mathcal{E}[B^3]\sin^3\omega t \\
&= \mathcal{E}[A^3](\cos^3\omega t + \sin^3\omega t) \\
&= \frac{\mathcal{E}[A^3]}{4}\{3\underbrace{(\cos\omega t + \sin\omega t) + (\cos 3\omega t - \sin 3\omega t)}_{\text{neither of these are constant}}\}
\end{aligned}$$

Third
$\Rightarrow$ ~~third~~ through moment of $X(t)$ depends on time
$\Rightarrow X(t)$ is not stationary
Since $\mathcal{E}[A^2B] = \mathcal{E}[A^2]\mathcal{E}[B] = \tau$ and $\mathcal{E}[AB^2] = 0$

$$\begin{aligned}
\cos^3\omega t &= \frac{1}{4}\cos 3\omega t + \frac{3}{4}\cos\omega t \\
\sin^3\omega t &= \frac{3}{4}\sin\omega t - \frac{1}{4}\sin 3\omega t
\end{aligned}$$

9.66

~~9.54~~ Assume X_n is discrete-valued, for simplicity, so that we can work with pmf's. Consider the third-order joint pmf of Y_n: for $n_1 < n_2 < n_3$ we need to show that for all $\tau > 0$

$$(\star)\quad P[Y_{n_1} = y_1, Y_{n_2} = y_2, Y_{n_3} = y_3] = P[Y_{n_1+\tau} = y_1, Y_{n_2+\tau} = y_2, Y_{n_3+\tau} = y_3]$$

Express the above probabilities in terms of the X_n's:

$$\begin{aligned}
&P[Y_{n_1} = y_1, Y_{n_2} = y_2, Y_{n_3} = y_3] \\
&\quad = P\left[\frac{1}{2}(X_{n_1} + X_{n_1-1}) = y_1, \frac{1}{2}(X_{n_2} + X_{n_2-1}) = y_2, \frac{1}{2}(X_{n_3} + X_{n_3-1}) = y_3\right] \\
&\quad = P\left[\frac{1}{2}(X_2 + X_1) = y_1, \frac{1}{2}(X_{n_2-n_1+2} + X_{n_2-n_1+1}) = y_2,\right. \\
&\quad\quad \left.\frac{1}{2}(X_{n_3-n_1+2} + X_{n_3-n_1+1}) = y_3\right]
\end{aligned}$$

Since the joint pdf of $(X_{n_1-1}, X_{n_1}, X_{n_2-1}, X_{n_2}, X_{n_3-1}, X_{n_3})$ is identical to that of $(X_1, X_2, X_{n_2-n_1+1}, X_{n_2-n_1+2}, \ldots, X_{n_3-n_1+2})$ if X_n is a stationary process.

Similarly we have that

$$\begin{aligned}
&P[Y_{n_1+\tau} = y_1, Y_{n_2+\tau} = y_2, Y_{n_3+\tau} = y_3] \\
&\quad = P\left[\frac{1}{2}(X_{n_1+\tau} + X_{n_1+\tau-1}) = y_1, \ldots \frac{1}{2}(X_{n_3+\tau} + X_{n_3+\tau-1}) = y_3\right] \\
&\quad = P\left[\frac{1}{2}(X_2 + X_1) = y_1, \frac{1}{2}(X_{n_2-n_1+2} + X_{n_2-n_1+1}) = y_2, \frac{1}{2}(X_{n_3-n_1+2} + X_{n_3-n_1+1}) = y_3\right]
\end{aligned}$$

$\therefore$ $(\star)$ holds if X_n is a stationary random process and in particular if X_n is an iid process. (a)

(b) — continued —

(c)

In order for X_n and Y_n to be jointly stationary, their joint distributions should be invariant to shifts in the origin.

In parts (a) and (b) we express the joint pmf of Y_n's in terms of the X_n's. Therefore we can also express joint pmf's of X_n's and Y_n's in terms of joint pmf's of X_n's only. The stationarity of X_n then implies shift invariance for the joint pmf's of X_n and Y_n.

$\therefore$ X_n and Y_n are jointly stationary if X_n is iid and if X_n is stationary.

9.67

6.55 $Z_n = \frac{3}{4}Z_{n-1} + X_n \qquad Z_0 = 0$

a) $(\star)\ Z_n = \sum_{i=1}^{n} \left(\frac{3}{4}\right)^{n-i} X_i \qquad \mathcal{E}[Z_n] = 0$

$m < n$

$$\begin{aligned} C_Z(m,n) &= \mathcal{E}[Z_m Z_n] = \mathcal{E}\left[\sum_{i=1}^{m}\left(\frac{3}{4}\right)^{m-i} X_i \sum_{j=1}^{n}\left(\frac{3}{4}\right)^{n-j} X_j\right] \\ &= \sum_{i=1}^{m}\sum_{j=1}^{n}\left(\frac{3}{4}\right)^{m+n-i-j}\mathcal{E}[X_i X_j] \\ &= \sum_{i=1}^{m}\left(\frac{3}{4}\right)^{m+n-2i}\mathcal{E}[X^2] \\ &\quad \text{since } \mathcal{E}[X_i X_j] = \begin{cases} 0 & i \neq j \\ \mathcal{E}[X^2] & i = j \end{cases} \\ &= \left(\frac{3}{4}\right)^{m+n}\mathcal{E}[X^2]\sum_{i=1}^{m}\left(\frac{16}{9}\right)^{i} \\ &= \left(\frac{3}{4}\right)^{m+n}\mathcal{E}[X^2]\frac{1-\left(\frac{16}{9}\right)^{m}}{1-\frac{16}{9}} = \left(\frac{1}{7}\right)^{2m} \end{aligned}$$

$$= \frac{9}{7}\mathcal{E}[X^2]\left[\underbrace{\left(\frac{3}{4}\right)^{n-m}}_{\substack{\text{dependence} \\ \text{on } n-m}} - \underbrace{\left(\frac{3}{4}\right)^{m+n}}_{\substack{\text{dependence} \\ \text{on } m,n}}\right]$$

$\Rightarrow Z_n$ is <u>not</u> WSS.

b) For τ a fixed time shift, $m, n = m + \tau$ as $m \to \infty$

$$C_Z(m, m+\tau) = \frac{9}{7}\mathcal{E}[X^2]\left(\frac{3}{4}\right)^{\tau}$$

$\therefore Z_n$ is asymptotically WSS.

Indeed as $m \to \infty$, we can suppose that the process stated at $t = -\infty$, then

$$Z_n = X_n + \frac{3}{4}X_{n-1} + \left(\frac{3}{4}\right)^2 X_{n-2} + \ldots$$

If X_n is a stationary process (and hence its joint pmf's are invariant with respect to time shifts) then Z_n will also be <u>stationary</u>.

c) From part a), $\mathcal{E}[Z_n] = 0$, and from part b)

$$C_Z(m, m+\tau) = \frac{9}{7}(1)\left(\frac{3}{4}\right)^{\tau} \qquad \text{as } m \to \infty.$$

Z_n is then a discrete-time, zero-mean, Gaussian random process with the above covariance function.

9.68

$Y(t) = X(t+s) - \beta X(t)$ $\quad E[X(t)] = m_X$, $\quad C_X(t_1,t_2) = C_X(t_1-t_2) = C_X(\tau)$

a) $E[Y(t)] = E[X(t+s)] - \beta E[X(t)] = m_X - \beta m_X = m_X(1-\beta)$

$$R_Y(t_1,t_2) = E[Y(t_1)Y(t_2)] = E[X(t_1+s)X_2(t_2+s)] - \beta^2 E[X(t_1)X(t_2)]$$

$$- \beta E[X(t_1+s)X(t_2)] - \beta E[X(t_2+s)X(t_1)]$$

$$= R_X(t_1-t_2) - \beta^2 R_X(t_2-t_2) - \beta R_X(t_1+s-t_2) - \beta R_X(t_2+s-t_1)$$

$$= (1-\beta^2) R_X(\tau) - \beta (R_X(\tau+s) + R_X(s-\tau)) \qquad \tau = t_1 - t_2$$

if s is fixed then $R_Y(t_1,t_2)$ only depends on τ.

Also, $C_Y(t_1,t_2) = R_Y(t_1,t_2) - m_Y(t_1)m_Y(t_2)$ only depends on t_1-t_2

Therefore Y is WSS

b) $C_{XY}(t_1,t_2) = E[X(t_1)Y(t_2)] - E[X(t_1)]E[Y(t_2)]$

$$= E[X(t_1)X(t_2+s)] - \beta E[X(t_1)X(t_2)] - m_X^2(1-\beta)$$

$$= R_X(t_1-t_2-s) - \beta R_X(t_1-t_2) - m_X^2(1-\beta)$$

again for fixed s, X & Y are jointly WSS

c) & d) $Y(t)$ have jointly Gaussian pdf's with mean m_X and covariance $C_X(\tau)$.

e) $X(t)$ and $Y(t)$ have jointly Gaussian pdf's with means m_X, m_Y, and covariances $C_X(\tau)$, $C_Y(\tau)$ and $C_{XY}(\tau)$.

9.69 $Z(t) = 3X(t) - 5Y(t)$

X & Y are WSS then $m_X = m_Y = 0$ & $\sigma_X^2 = \sigma_Y^2 = C_X(0)$

a) $m_Z(t) = 3m_X(t) - 5m_Y(t) = 0$

$$C_Z(t_1,t_2) = R_Z(t_1,t_2) - 0 = R_Z(t_1,t_2) =$$

$$= E\{(3X(t_1) - 5Y(t_1))(3X(t_2) - 5Y(t_2))\}$$

$$= 9C_X(t_1,t_2) + 25C_Y(t_1,t_2) - 15C_{XY}(t_1,t_2) - 15C_{YX}(t_1,t_2)$$

X & Y independent $= 9C_X(\tau) + 25C_Y(\tau)$

Then Z is WSS

b) $Z(t)$ would be a Gaussian RV. with zero mean

and $\sigma_{Z(t)}^2 = 34\sigma_{X(t)}^2$ $\sigma_{Z(t)}^2 = 34\sigma_{X(t)}^2 = 34\,C_X(0) = 34 \times 4 = 132$

$$f_{Z(t)}(z) = \frac{1}{\sqrt{2\pi}\,\sigma_{Z(t)}} e^{\frac{-(z)^2}{2\sigma_{Z(t)}^2}}$$

c) $Z(t_1) = 3X(t_1) - 5Y(t_1)$ & $Z(t_2) = 3X(t_2) - 5Y(t_2)$

$Z(t_1)$ & $Z(t_2)$ are both Gaussian RVs. with mean $m_{Z(t_1)}$ & $m_{Z(t_2)}$ and variance $\sigma_{Z(t_1)}^2$ & $\sigma_{Z(t_2)}^2$, Also $C_Z(t_1,t_2) = C_Z(t_1 - t_2) = 132\,e^{-|t_1 - t_2|}$

So $f_{Z(t_1)Z(t_2)}(z_1,z_2)$ can be obtained as in P9.51

9.70

~~6.58~~ **a)** From Problem ~~6.18~~ 9.55:

$$\begin{aligned}\mathcal{E}[Z(t)] &= 0\\ C_Z(t_1,t_2) &= C_X(t_2-t_1)\cos\omega(t_2-t_1)\end{aligned}$$

$\Rightarrow Z(t)$ is WSS

b) $Z(t)$ is a Gaussian RV with mean zero and variance $C_X(0)$.

(c) $Z(t)$ is WSS Gaussian RP with zero mean and covariance $C_X(t_2-t_1)\cos\omega(t_2-t_1)$

$\therefore$ $(Z(t_1), Z(t_2))$ has jointly Gaussian pdf with

$$\underline{m}_X = \begin{bmatrix}0\\0\end{bmatrix} \qquad \underline{K}_X = \begin{bmatrix} C_X(0) & C_X(t_2-t_1)\cos\omega(t_2-t_1) \\ C_X(t_2-t_1)\cos\omega(t_2-t_1) & C_X(0)\end{bmatrix}$$

(d) Since processes are zero mean

$$\begin{aligned}\text{Cov}(Z(t_1)X(t_2)) &= E[Z(t_1)X(t_2)]\\ &= [(X(t_1)\cos\omega t_1 + Y(t_1)\sin\omega t_1)X(t_2)]\\ &= E[X(t_1)X(t_2)]\cos\omega t_1\\ &= R_X(t_2-t_1)\cos\omega t_1\end{aligned}$$

dependence on t_1

(9) Use auxiliary variables

$$Z(t_1) = X(t_1)\cos\omega t_1 + Y(t_1)\sin\omega t_1$$

$$W(t_2) = X(t_2)$$

$$V(t_2) = Y(t_2)$$

$$\begin{bmatrix} Z \\ W \\ V \end{bmatrix} = \begin{bmatrix} \cos\omega t_1 & 0 & \sin\omega t_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ Y_1 \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \\ Y_1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\cos} & 0 & -\frac{\sin}{\cos} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Z \\ W \\ V \end{bmatrix}$$

The joint pdf of $X(t)$ and $Z(t)$ have parameters

$$m_X(t) = 0, \quad C_X(t_1 - t_2)$$

$$m_Z(t) = 0, \quad C_Z(t_1, t_2) = C_X(t_2 - t_1)\cos\omega(t_2 - t_1)$$

and $$\text{cov}(Z(t_1), X(t_2)) = R_X(t_2 - t_1)\cos\omega t_1$$

$$\underline{m}_{XZ} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad K_{X,Z} = \begin{bmatrix} C_X(t_2 - t_1) & R_X(t_2 - t_1)\cos\omega t_1 \\ R_X(t_2 - t_1)\cos\omega t_1 & C_X(t_2 - t_1)\cos\omega(t_2 - t_1) \end{bmatrix}$$

9.71

$$R_Y(t_1,t_2) = E[Y(t_1)Y(t_2)] = E[X^2(t_1)X^2(t_2)] =$$

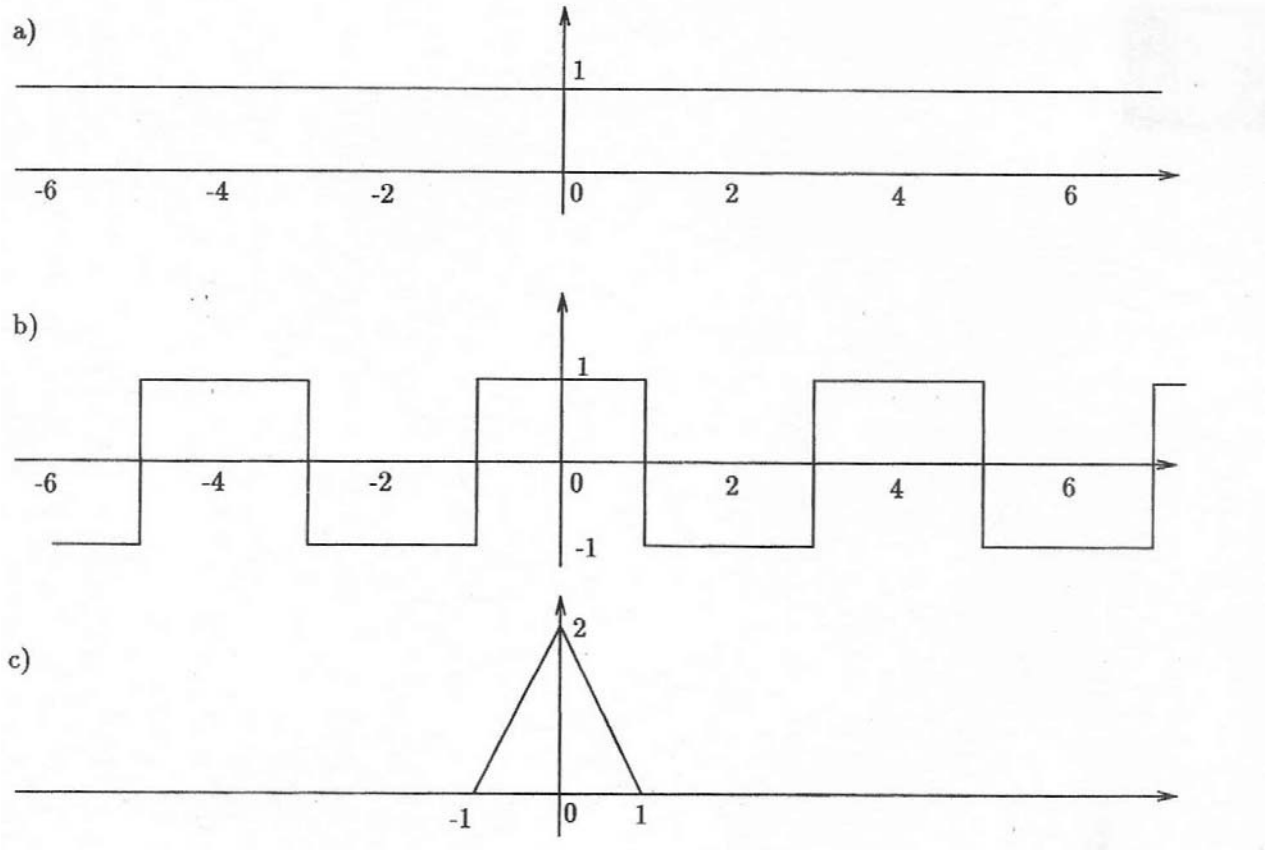
$X(t_1)$ & $X(t_2)$ are zero mean jointly Gaussian RV

Hint: $$R_Y(t_1,t_2) = E[X^2(t_1)]E[X^2(t_2)] + 2E[X(t_1)X(t_2)]^2$$

$$= R_X(0)R_X(0) + 2R_X^2(\tau), \quad \tau = t_1 - t_2$$

$$\Rightarrow \quad R_Y(\tau) = R_X^2(0) + 2R_X^2(\tau)$$

9.72



9.73 The sequence U_n and the sequences X_n and Y_n are related as shown below:

$$\cdots \mid \underline{U_{-2}} \quad \underline{U_{-1}} \mid U_0 \quad U_1 \mid U_2 \quad U_3 \mid U_4 \quad U_5 \mid \cdots$$
$$\cdots \mid \underline{X_{-1}} \quad \underline{Y_{-1}} \mid X_0 \quad Y_0 \mid X_1 \quad Y_1 \mid X_2 \quad Y_2 \mid \cdots$$

If the sequence U_n is shifted by $2k$, then the subsequences X_n and Y_n are shifted by k.

If the X_n (Y_n) are stationary then their joint pmf's are shift invariant.

(b) $\therefore$ the joint pmf of U_n is also shift invariant and U_n is cyclostationary.

(a) If X_n and Y_n are iid then they are stationary $\Rightarrow U_n$ is cyclostationary.

(c) If X_n and Y_n are WSS then m_X = constant m_Y = constant $R_X(\tau)$ and $R_Y(\tau)$ depend on $t_2 - t_1$ only.

Consider a shift of $2k$, then

$$m_U(t+2k) = m_U(t) = \begin{cases} m_X & t \text{ even} \\ m_Y & t \text{ odd} \end{cases}$$

also

$$C_U(t_1+2k, t_2+2k) = C_U(t_1, t_2) = \begin{cases} C_X(t_1,t_2) & t_1+t_2 \text{ even} \\ C_Y(t_1,t_2) & t_1+t_2 \text{ odd} \\ C_{XY}(t_1,t_2) & t_1 \text{ even}, t_2 \text{ odd} \\ C_{YX}(t_1,t_2) & t_1 \text{ odd}, t_2 \text{ even} \end{cases}$$

$\therefore U_n$ is wide sense cyclostationary

In general U_n is NOT WSS since X and Y may have different mean and covariance functions.

(d) $m_U = \frac{1}{2} m_X + \frac{1}{2} m_Y$

$$C_U(t_1, t_2) = \frac{1}{2} C_X(t_1, t_2) + \frac{1}{2} C_Y(t_1, t_2)$$
$$= \frac{1}{2} C_X(t_2 - t_1) + \frac{1}{2} C_Y(t_2 - t_1)$$

9.74

6.62 a) If n is even

$$\begin{aligned} P[B_n = 0, B_{n+1} = 0] &= \frac{1}{3} \\ P[B_n = 0, B_{n+1} = 1] &= \frac{1}{3} \\ P[B_n = 1, B_{n+1} = 0] &= \frac{1}{3} \\ P[B_n = 1, B_{n+1} = 1] &= 0 \end{aligned}$$

If n is odd

$$\begin{aligned} P[B_n = 0, B_{n+1} = 0] &= \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9} \\ P[B_n = 0, B_{n+1} = 1] &= \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9} \\ P[B_n = 1, B_{n+1} = 0] &= \frac{2}{9} \\ P[B_n = 1, B_{n+1} = 1] &= \frac{1}{9} \end{aligned}$$

B_n is not stationary, but is cyclostationary with period 2.

b) $E[B_n] = 0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \frac{1}{3}$.

If n is even

$$\begin{aligned} R_B(n, n+j) &= E[B_n B_{n+j}] = P[B_n = 1, B_{n+j} = 1] \\ &= \begin{cases} 0 & \text{if } j = 1 \\ \frac{1}{3} & \text{if } j = 0 \\ \frac{1}{9} & \text{otherwise} \end{cases} \end{aligned}$$

if n is odd,

$$\begin{aligned} R_B(n, n+j) &= E[B_n B_{n+j}] = P[B_n = 1, B_{n+j} = 1] \\ &= \begin{cases} 0 & \text{if } j = -1 \\ \frac{1}{3} & \text{if } j = 0 \\ \frac{1}{9} & \text{otherwise} \end{cases} \end{aligned}$$

so B_n is not wide-sense stationary, but wide-sense cyclostationary.

c) After introducing random phase, we obtain B_n^s.

$$P[B_n^s = 0] = \frac{2}{3}, \qquad P[B_n^s = 1] = \frac{1}{3}$$

$$\begin{aligned} P[B_n^s = 0, B_{n+1}^s = 0] &= P[B_n = 0, B_{n+1} = 0 | n \text{ even}]\frac{1}{2} + P[B_n = 0, B_{n+1} = 0 | n \text{ odd}]\frac{1}{2} \\ &= \frac{1}{2}\cdot\frac{1}{3} + \frac{1}{2}\cdot\frac{4}{9} = \frac{7}{18} \end{aligned}$$

Similarly

$$\begin{aligned} P[B_n^s = 0, B_{n+1}^s = 1] &= \frac{1}{2}\cdot\frac{1}{3} + \frac{1}{2}\cdot\frac{2}{9} = \frac{5}{18} \\ P[B_n^s = 1, B_{n+1}^s = 0] &= \frac{5}{18} \\ P[B_n^s = 1, B_{n+1} = 1] &= \frac{1}{2}\frac{1}{9} = \frac{1}{18} \end{aligned}$$

$$E[B_n^s] = \frac{1}{3}$$

$$R_{B^s}(n, n+j) = \begin{cases} \frac{1}{3} & j = 0 \\ \frac{1}{18} & j = \pm 1 \\ \frac{1}{9} & \text{otherwise} \end{cases}$$

$$C_{B_s}(n, n+j) = \begin{cases} \frac{2}{9} & j = 0 \\ -\frac{1}{18} & j = \pm 1 \\ 0 & \text{otherwise} \end{cases}$$

9.75

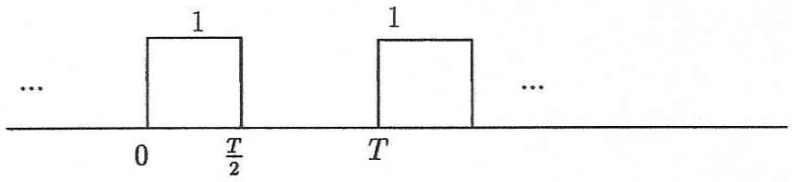
~~6.63~~ a)
$$\begin{aligned} E[X(t)] &= E[As(t)] = E[A]s(t) \\ R_X(t_1,t_2) &= E[As(t_1)As(t_2)] = E[A^2]s(t_1)s(t_2) \\ C_X(t_1,t_2) &= R_X(t_1,t_2) - E[X(t_1)]E[X(t_2)] = VAR[A]s(t_1)s(t_2) \end{aligned}$$

X(t) is not WSS and hence not mean-square periodic.

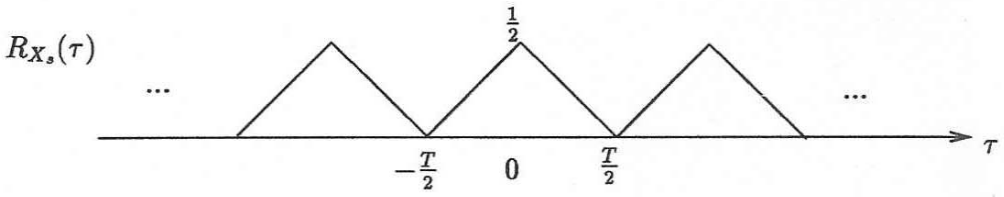
b)
$$\begin{aligned} X_s(t) &= X(t+\theta) \\ E[X_s(t)] &= \frac{1}{T}\int_0^T m_X(t)dt \end{aligned}$$

$$\begin{aligned} &= \frac{1}{T}\int_0^T E[A]s(t)dt = E[A]\frac{1}{T}\int_0^T s(t)dt \\ R_{X_s}(\tau) &= \frac{1}{T}\int_0^T R_X(t+\tau,t)dt \\ &= \frac{1}{T}\int_0^T E[A^2]s(t+\tau)s(t)st \\ &= E[A^2]\frac{1}{T}\int_0^T s(t+\tau)s(t)dt \end{aligned}$$

Thus the mean and autocorrelation of $X_s(t)$ are determined by time averages of $s(t)$. If $s(t)$ is as below



then
$$\frac{1}{T}\int_0^T s(t)dt = \frac{1}{2} \quad \text{and} \quad \frac{1}{T}\int_0^T s(t)s(t-\tau)dt = \frac{1}{2} - \frac{\tau}{T} \quad |\tau| < \frac{T}{2}$$



9.76)

a) if $A = 1$, $X(t_1) = s(t_1)$

if $A = -1$, $X(t_1) = -s(t_1)$

$$P\{X(t_1)=1\} = \frac{1}{2} \times P\{s(t_1)=1\} + \frac{1}{2} \times P\{s(t_1)=-1\} = \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2} = \frac{1}{2}$$

Also, similarly $P\{X(t_1)=-1\} = \frac{1}{2}$, $P\{Y(t_2)=1\} = \frac{1}{2}$, $P\{Y(t_2)=-1\} = \frac{1}{2}$

$$P\{X(t_1)=\pm 1, Y(t_2)=\pm 1\} = P\{X(t_1)=\pm 1\}\,P\{Y(t_2)=\pm 1\} = \frac{1}{4}$$

distributed

note that since A & B are independent, identically RVs, then X & Y become independent for all t.

If $\{\cdot\}$ is indicator function

since one and only one of the indicator functions is 1.

$$P\{X(t_1)=1, Y(t_2)=1\} = P\{A=1, B=1\}\,\mathbf{I}\{s(t_1)=1, s(t_2)=1\}$$
$$+ P\{A=1, B=-1\}\,\mathbf{I}\{s(t_1)=+1, s(t_2)=-1\}$$
$$+ P\{A=-1, B=1\}\,\mathbf{I}\{s(t_1)=-1, s(t_2)=1\}$$
$$+ P\{A=-1, B=-1\}\,\mathbf{I}\{s(t_1)=-1, s(t_2)=-1\}$$
$$= \frac{1}{4} = P\{X(t_1)=1\} \times P\{Y(t_2)=1\} \quad \forall t_1, t_2$$

This can be shown for any value of $X(t_1)$ & $Y(t_2)$, and for n-th order pmf

b) $C_{XY}(t_1,t_2) = E[X(t_1)Y(t_2)] - E[X(t_1)]E[Y(t_2)]$

$E[X(t_1)] = E[Y(t_2)] = 0$

therefore $C_{XY}(t_1,t_2) = E[X(t_1)Y(t_2)] = \frac{1}{4}\times 1 + \frac{1}{4}\times -1 + \frac{1}{4}\times -1 + \frac{1}{4}\times 1 = 0 \quad \forall t_1, t_2$

which means X & Y are uncorrelated.

c) $C_{XY}(t_1+mT, t_2+mT) = \emptyset = C_{XY}(t_1,t_2)$; and they are independent so their n-th order pmf is also

⇒ jointly WS cyclostationary

X_n and Y_n

jointly cyclostationary.

9.77

6.64 Recall the application of the Schwarz Inequality (Eqn. 6.60) during the discussion after the mean square periodic process was defined on page 360. We had:

$$E[(X(t+\tau+d)-X(t+\tau)X(t))]^2 \leq E[(X(t+\tau+d)-X(t+\tau))^2]E[X^2(t)]$$

If $X(t)$ is mean-square periodic, then

$$E[(X(t+\tau+d)-X(t+d))^2]=0 \quad .$$

Thus

$$\begin{aligned} & E[(X(t+\tau+d)-X(t+\tau))X(t)]^2=0 \\ \Rightarrow \quad & (E[X(t+\tau+d)X(t)]-E[X(t+\tau)X(t)])^2=0 \\ \Rightarrow \quad & E[X(t+\tau+d)X(t)]=E[X(t+\tau)X(t)] \\ \Rightarrow \quad & R_X(t_1+d,t_2)=R_X(t_1,t_2) \end{aligned}$$

Repeated applications of this argument to t_1 and t_2 implies

$$R_X(t_1+md,t_2+nd)=R_X(t_1,t_2) \quad \text{for every integer } m,n \ .$$

The special case $m=n$ implies (6.63b) and hence that $X(t)$ is wide-sense cyclostationary.

9.78

6.65 Since the data sequence is iid, each T-second interval can be viewed as an independent trial. Therefore, time shifts of the process by integer multiples of T leave the joint distributions unchanged. Thus the process is cyclostationary.

9.79

6.66

$$\begin{aligned} m_X(t) &= E[A]\cos\frac{2\pi t}{T} \\ E[X_S(t)] &= \frac{1}{T}\int_0^T E[A]\cos\frac{2\pi t}{T}dt=0 \\ R_X(t,t+\tau) &= E\left[A\cos\frac{2\pi t}{T}A\cos\frac{2\pi(t+\tau)}{T}\right] \\ &= E[A^2]\frac{1}{2}\left(\cos\frac{2\pi\tau}{T}+\cos\frac{4\pi t+2\pi\tau}{T}\right) \\ R_{X_S}(\tau) &= \frac{1}{T}\int_0^T R_X(t,t+\tau)dt \\ &= \frac{1}{2}\cos\frac{2\pi\tau}{T}E[A^2] \end{aligned}$$

9.80

6.67 Note that $\int_{-\tau}^{T-\tau} f(t)dt = \int_0^T f(t)dt$ if $f(t+mT) = f(t)$

$$\begin{aligned}
&P[X_S(t_1+\tau) \le x_1, ..., X_S(t_k+\tau) \le x_k] \\
&\quad = P[X(t_1+\tau+\Theta) \le x_1, ..., X(t_k+\tau+\Theta) \le x_k] \\
&\quad = \int_0^T P[X(t_1+\tau+\Theta) \le x_1, ..., X(t_k+\tau+\Theta) \le x_k | \Theta = 0] d\Theta \\
&\quad \frac{1}{T}\int_{-\tau}^{T-\tau} P[X(t_1+\tau+\Theta) \le x_1, ..., X(t_k+\tau+\Theta) \le x_k] d\Theta \\
&\quad = \frac{1}{T}\int_0^T P[X(t_1+\Theta) \le x_1, ..., X(t_k+\Theta) \le x_k] d\Theta \\
&= P[X_S(t_1) \le x_1, ..., X_S(t_k) \le x_k] \\
&\quad \therefore X_S(t) \text{ is stationary}
\end{aligned}$$

9.81

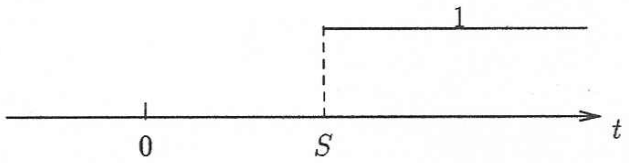
6.68 $X_S(t) = X(t+\Theta)$

$$\begin{aligned}
E[X_S(t)] &= E[X(t+\Theta)] \\
&= E[E[X(t+\Theta)|\Theta]] \\
&= E[m_X(t+\Theta)] \\
&= \frac{1}{T}\int_0^T m_X(t+\Theta)d\Theta \\
&= \frac{1}{T}\int_{+t}^{T+t} m_X(u)du \\
&= \frac{1}{T}\int_0^T m_X(u)du \\
R_{X_S}(\tau) &= E[X(t_1+\Theta)X(t_1+\tau+\Theta)] \\
&= E[E[X(t_1+\Theta)X(t_1+\tau+\Theta)|\Theta]] \\
&= E[R_X(t_1+\Theta, t_1+\tau+\Theta)] \\
&= \frac{1}{T}\int_0^T R_X(t_1+\Theta, t_1+\tau+\Theta)d\Theta \\
&= \frac{1}{T}\int_{+t_1}^{T+t_1} R_X(u, u+\tau)du \\
&= \frac{1}{T}\int_0^T F_X(u, u+\tau)du
\end{aligned}$$

$\therefore$ mean of $X_S(t)$ is constant, and autocorrelation depends only on τ.

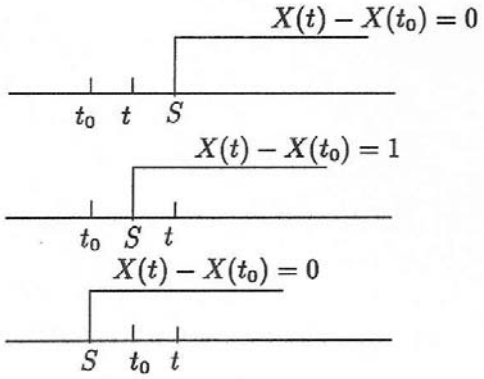
9.7 Continuity, Derivatives, and Integrals of Random Processes

9.82



a) $P[X(t) \text{ discontinuous at } t_0] = P[s = t_0] = 0$

b)



$$\begin{aligned}\lim_{t_0 \to 0} E[X(t) - X(t_0)^2] &= 1 \cdot P[t_0 < S < t] \\ &= e^{-\lambda t} - e^{-\lambda t_0} \\ &\to 0 \quad \Rightarrow \quad X(t) \text{ is M.S. continuous}\end{aligned}$$

We can also determine continuity from the autocorrelation function:

$$\begin{aligned}E[X(t_1)X(t_2)] &= P[X > \max(t_1, t_2)] \\ &= e^{-\lambda \max(t_1, t_2)}\end{aligned}$$

Next we determine if $R_X(t_1, t_2)$ is continuous at (t_0, t_0):

$$\begin{aligned}R_X(t_0 + \varepsilon_1, t_0 + \varepsilon_2) - R_X(t_0, t_0) &= e^{-\lambda \max(t_0+\varepsilon_2, t_0+\varepsilon_2)} - e^{-\lambda t_0} \\ &= e^{-\lambda(t_0 + \max(\varepsilon_1, \varepsilon_2))} - e^{-\lambda t_0} \\ &= e^{-\lambda t_0}[e^{-\lambda \max(\varepsilon_1, \varepsilon_2)} - 1] \\ &\to 0 \quad \text{as } \varepsilon_1 \text{ and } \varepsilon_2 \to 0\,. \\ &\quad \Rightarrow \quad X(t) \text{ is M.S. continuous}\end{aligned}$$

c) We expect that the mean square derivative is zero (if it exists). We thus consider the limit:

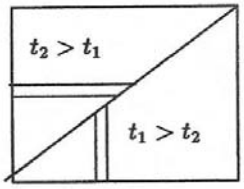
$$E\left[\left(\frac{X(t+\tau) - X(t)}{\varepsilon}\right)^2\right] = \frac{1}{\varepsilon^2}(e^{-\lambda(t+\varepsilon)} - e^{-\lambda t})$$

$$= e^{-\lambda t} \underbrace{\left(\frac{e^{-\lambda\varepsilon}-1}{\varepsilon^2}\right)}_{\frac{\lambda\varepsilon - \frac{\lambda^2\varepsilon^2}{2}+\ldots}{\varepsilon^2}}$$

$$= e^{-\lambda t}\frac{1}{\varepsilon} \to \infty \ .$$

Thus the M.S. derivative does not exist.

d) $X(t)$ is M.S. integrable if the following integral exists:



Two regions of integration

$$\begin{aligned}
&\int_0^t \int_0^t e^{-\lambda \max(t_1,t_2)} dt_1 dt_2 \\
&= \int_0^t dt_1 \int_0^{t_1} dt_2 e^{-\lambda t_1} + \int_0^t dt_2 \int_0^{t_2} dt_1 e^{-\lambda t_2} \\
&= \int_0^t dt_1 t_1 e^{-\lambda t_1} + \int_0^{t_2} dt_2 t_2 e^{-\lambda t_2} \\
&= \frac{e^{-\lambda t_1}(-\lambda t_1 - 1)}{\lambda^2}\Bigg|_0^t + \frac{e^{-\lambda t_2}(-\lambda t_2 - 1)}{\lambda^2}\Bigg|_0^t \\
&= \frac{e^{-\lambda t}(-\lambda t - 1) - (-1)}{\lambda^2} + \frac{e^{-\lambda t}(-\lambda t - 1) - (-1)}{\lambda^2} \\
&= \frac{2}{\lambda^2}[1 - e^{-\lambda t}(\lambda t + 1)] \\
&\Rightarrow \quad X(t) \text{ is M.S. integrable}
\end{aligned}$$

Let

$$Y(t) = \int_0^t X(\lambda) d\lambda \ .$$

Then from Eqn. 9.91

$$\begin{aligned}
m_Y(t) &= \int_0^t m_X(u) du \\
m_X(t) &= E[X(t)] = 1 \cdot P[S < t] = 1 - e^{-\lambda t} \\
m_Y(t) &= \int_0^t (1 - e^{-\lambda u}) du = t - \left(\frac{e^{-\lambda u}}{-\lambda}\right)_0^t \\
&= t + \frac{1}{\lambda}[e^{-\lambda t} - 1]
\end{aligned}$$

From Eqn. 9.92 we have:

$$\begin{aligned}
R_Y(t_1, t_2) &= \int_0^{t_1} \int_0^{t_2} R_X(u, v) du dv \\
&= \frac{1 - e^{-\lambda t_1}(\lambda t_1 + 1)}{\lambda^2} + \frac{1 - e^{-\lambda t_2}(\lambda t_2 + 1)}{\lambda^2}
\end{aligned}$$

9.83

6.70 a) $R_X(t_1,t_2) = e^{-2\alpha|t_1+t_2|}$

$R_X(t_1,t_2)$ is continuous in both t_1 and t_2.

$X(t)$ is M.S. continuous.

b)
$$R_X(t_1,t_2) = \begin{cases} e^{-2\alpha(t_1-t_2)} & t_2 < t_1 \\ e^{-2\alpha(t_2-t_1)} & t_2 \geq t_1 \end{cases}$$

$$\frac{\partial R_X(t_1,t_2)}{\partial t_2} = \begin{cases} 2\alpha e^{-2\alpha(t_1-t_2)} & t_2 < t_1 \\ -2\alpha e^{-2\alpha(t_2-t_1)} & t_2 \geq t_1 \end{cases}$$

$$= sgn(t_1-t_2)2\alpha e^{-2\alpha|t_1-t_2|}$$

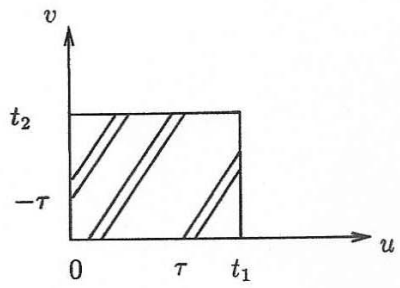
The second derivative does not exist at discontinuity points. $X(t)$ does not have a mean square derivative.

$$\frac{\partial^2 R_X(t_1,t_2)}{\partial t_1 \partial t_2} = \delta(t_1-t_2)4\alpha e^{-2\alpha(t_1-t_2)} + 4\alpha^2 e^{-2\alpha(t_1-t_2)}$$

The transitions from $+1$ to -1 or from -1 to 1 give rise to delta function in $R_{X'}(t_1,t_2)$.

c) $X(t)$ has a M.S. integral since it is M.S. continuous. $m_X(t) = 0$, so

$$E\left[\int_0^t x(t')dt'\right] = \int_0^t E[x(t')]dt' = 0$$



Consider $Y(t) = \int_0^t X(t')dt$

$$\begin{aligned}
R_Y(t_1,t_2) &= \int_0^{t_1}\int_0^{t_2} R_X(u,v)dudv \qquad (t_2, \le t_1) \\
&= \int_0^{t_1}\int_0^{t_2} R_X(u-v)dudv \\
&= \int_{-t_2}^{0}(t_2+\tau)R_X(\tau)d\tau + \int_0^{t_1-t_2} t_2 R_X(\tau)d\tau \\
&\quad + \int_{t_1-t_2}^{t_1}(t_1-\tau)R_X(\tau)d\tau \\
&= \int_{-t_2}^{0} e^{-2\alpha(-\tau)}(t_2+\tau)d\tau + \int_0^{t_1-t_2} t_2 e^{-2\alpha\tau}d\tau \\
&\quad + \int_{t_1-t_2}^{t_1}(t_1-\tau)e^{-2\alpha\tau}d\tau \\
&= \frac{t_2}{2\alpha}(1-e^{-2\alpha t_2}) + \frac{1}{2\alpha}\int_{-t_2}^{0}\tau de^{2\alpha\tau} + \frac{t_2}{2\alpha}(1-e^{-2\alpha(t_1-t_2)}) \\
&\quad + \frac{t_1}{2\alpha}(e^{-2\alpha(t_1-t_2)} - e^{-2\alpha t_1}) + \frac{1}{2\alpha}\int_{t_1-t_2}^{t_1}\tau de^{-2\alpha\tau} \\
&= \frac{t_2}{2\alpha}(1-e^{-2\alpha t_2}) + \frac{t_2}{2\alpha}e^{-2t_2} - \frac{1}{2\alpha}\int_{-t_2}^{0} e^{2\alpha\tau}d\tau + \frac{t_2}{2\alpha}(1-e^{-2\alpha(t_1-t_2)}) \\
&\quad + \frac{t_1}{2\alpha}(e^{-2\alpha(t_1-t_2)} - e^{-2\alpha t_1}) + \frac{1}{2\alpha}(t_1e^{-2\alpha t_1} - (t_1-t_2)e^{-2\alpha(t_1-t_2)}) \\
&\quad - \frac{1}{2\alpha}\int_{t_1-t_2}^{t_1} e^{-2\alpha\tau}d\tau \\
&= \frac{t_2}{2\alpha}(1-e^{-2\alpha t_2}) + \frac{t_2}{2\alpha} - \frac{1}{4\alpha^2}(1-e^{-2\alpha t_2}) + \frac{t_2}{2\alpha}(1-e^{-2\alpha(t_1-t_2)}) \\
&\quad + \frac{t_1}{2\alpha}(e^{-2\alpha(t_1-t_2)} - e^{-2\alpha t_1}) + \frac{1}{2\alpha}(t_1e^{-2\alpha t_1} - (t_1-t_2)e^{-2\alpha(t_1-t_2)}) \\
&\quad = \frac{1}{4\alpha^2}[e^{-2\alpha t_1} - e^{-2\alpha(t_1-t_2)}] \\
&= \frac{t_2}{2\alpha} - \frac{1}{4\alpha^2}(1-e^{-2\alpha t_2}) + \frac{t_2}{2\alpha}(1-2e^{-2\alpha(t_1-t_2)}) + \frac{1}{4\alpha^2}[e^{-2\alpha t_1} - e^{-2\alpha(t_1-t_2)}]
\end{aligned}$$

In general

$$\begin{aligned}
R_Y(t_1,t_2) &= \frac{\min(t_1,t_2)}{\alpha}(1-e^{-2\alpha(t_1-t_2)}) \\
&\quad + \frac{1}{4\alpha^2}(e^{-2\alpha t_1} + e^{-2\alpha t_2} - 1 - e^{-2\alpha(t_1-t_2)})
\end{aligned}$$

9.84

6.71 $R_X(\tau) = \sigma^2 e^{-\alpha\tau^2}, \quad R_X(t_1,t_2) = \sigma^2 e^{-\alpha(t_1-t_2)^2}$

a) Yes since $R_X(\tau)$ is continuous at τ.

b) Yes since $R_X(\tau)$ has derivatives of all orders at $\tau = 0$.

$$\begin{aligned} E\left[\frac{d}{dt}X(t)\right] &= \frac{d}{dt}[E[X(t)]] = 0, \quad \text{because } E[X(t)] = \text{constant} \\ R_{X'}(\tau) &= -\frac{d}{d\tau^2}R_X(\tau) \\ &= 2\alpha\sigma^2 e^{-\alpha\tau^2}(1-2\alpha\tau^2) \end{aligned}$$

c) Yes since $R_X(\tau)$ is M.S. continuous.
Consider $Y(t) = \int_0^t X(t)dt$.

$$\begin{aligned} E[Y(t)] &= \int_0^t E[X(t)]dt = m_X t \\ R_Y(t_1,t_2) &= \int_0^{t_1}\int_0^{t_2} R_X(u,v)dudv \end{aligned}$$

If $t_2 \le t_1$

$$\begin{aligned} R_Y(t_1,t_2) &= \int_0^{t_1}\int_0^{t_2} R_X(u-v)dudv \\ &= \int_{t_2}^{0}(t_2+\tau)\sigma^2 e^{-\alpha\tau^2}d\tau + \int_0^{t_1-t_2} t_2\sigma^2 e^{-\alpha\tau^2}d\tau \\ &\quad + \int_{t_1-t_2}^{t_1}(t_1-\tau)\sigma^2 e^{-v\tau^2}d\tau \\ &= \sigma^2 t_2\int_{-t_2}^{0} e^{-\alpha\tau^2}d\tau + \frac{\sigma^2}{2\alpha}\int_{-t_2}^{0} e^{-\alpha\tau^2}d\alpha\tau^2 + \sigma^2 t_2\int_0^{t_1-t_2} e^{-\alpha\tau^2}d\tau \\ &\quad + \sigma^2 t_1\int_{t_1-t_2}^{t_1} e^{-\alpha\tau^2}d\tau - \frac{\sigma^2}{2\alpha}\int_{t_1-t_2}^{t_1} e^{-\alpha\tau^2}d\sigma\tau^2 \\ &= \sigma^2 t_2\int_{-t_2}^{t_1-t_2} e^{-\alpha\tau^2}d\tau - \frac{\sigma^2}{2\alpha}(t - e^{-\alpha t_2^2}) \\ &\quad + \sigma^2 + 1\int_{t_1-t_2}^{t_1} e^{-\alpha\tau^2}d\tau + \frac{\sigma^2}{2\alpha}(e^{-\alpha t_1^2} - e^{-\alpha(t_1-t_2)^2}) \end{aligned}$$

If $t_2 > t_1$

$$\begin{aligned} R_Y(t_1,t_2) &= \sigma^2 t_1\int_{-t_1}^{t_2-t_1} e^{-\alpha\tau^2}d\tau + \sigma^2 t_2\int_{t_2-t_1}^{t_2} e^{-\alpha\tau^2}d\tau \\ &\quad + \frac{\sigma^2}{2\alpha}(-1 + e^{-\alpha t_1^2} + e^{-\alpha t_2^2} - e^{-\alpha(t_1-t_2)^2}) \end{aligned}$$

d) The fact that the autocorrelation has the shape of a Gaussian pdf does not imply that the process is Gaussian.

9.85

6.72 The independent increments property implies that

$$E[(N(t) - N(t_0))^2] = E[(N^2(t - t_0))] = \lambda(t - t_0) + \lambda^2(t - t_0)^2$$

So

$$\lim_{t \to t_0} E[(N(t) - N(t_0))^2] = 0$$

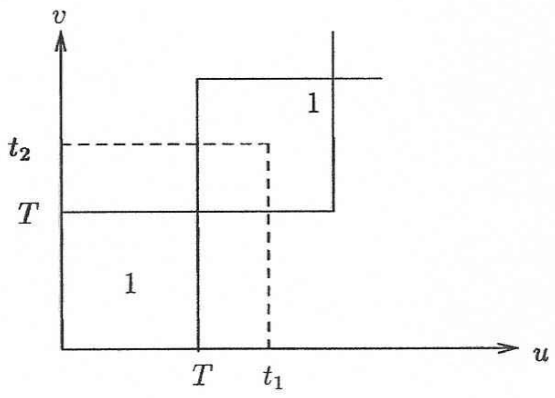
and $N(t)$ is M.S. continuous.

9.86

6.73 $X(t)$ is not M.S. continuous so we don't know whether it is integrable. Consider $Y(t) = \int_0^t X(t')dt'$, then

$$E[Y(t)] = 0$$

Now consider $$R_Y(t_1, t_2) = \int_0^{t_1} \int_0^{t_2} R_X(u, v)dudv$$



$R_Y(t_1, t_2)$ is the area of the non-zero region within the rectangle defined by (t_1, t_2). For fixed values t_1, t_2 this area is well-defined. Therefore $X(t)$ is M.S. integrable.

9.87

6.74 From ordinary calculus we know that the integral in Eqn. 6.83 exists if its argument $R_X(u,v)$ is a continuous function of u and v. Therefore, it suffices for us to show that if $X(t)$ is M.S. continuous, then $R_X(u,v)$ is continuous. Therefore consider

$$\begin{aligned} R_X(t_1,t_2) - R_X(t_0,t_0) &= E[X(t_1)X(t_2) - X(t_0)X(t_0)] \\ &= E[X(t_1)X(t_2) - X(t_0)X(t_2) + X(t_0)X(t_2) - X(t_0)X(t_0)] \\ &= E[(X(t_1) - X(t_0))X(t_2)] + E[X(t_0)(X(t_2) - X(t_0))] \end{aligned}$$

From the Schwarz Inequality (Eqn. 6.60)

$$|E[(X(t_1) - X(t_0))X(t_2)]|^2 \leq E[(X(t_1) - X(t_0))^2]E[X^2(t_2)]$$

and

$$|E[(X(t_0))(X(t_2) - X(t_0))]|^2 \leq E[X^2(t_0)]E[(X(t_2) - X(t_0))^2]$$

If $X(t)$ is M.S. continuous then the right-hand side approaches zero as $t_2 \to t_0$ and $t_1 \to t_0$. Therefore

$$R_X(t_1,t_2) \to R_X(t_0,t_0)$$

and hence $R_X(u,v)$ is continuous as required.

9.88

6.75 $Y(t) = \int_0^t X(u)du$

$$\begin{aligned} E\left[\left(\frac{d}{dt}Y(t) - X(t)\right)^2\right] &= E\left[\frac{d}{dt}Y(t)\frac{d}{dt}Y(t)\right] - 2E\left[X(t)\frac{d}{dt}Y(t)\right] \\ &\quad + E[X(t)X(t)] \\ &= \frac{\partial^2}{\partial t_1 \partial t_2}R_Y(t_1,t_2)\bigg|_{t_1=t_2=t} - 2\left.\frac{\partial R_{XY}(t_1,t_2)}{\partial t_2}\right|_{t_1=t_2=t} + R_X(t,t) \end{aligned}$$

$$\begin{aligned} R_Y(t_1,t_2) &= \int_0^{t_1}\int_0^{t_2} R_X(u,v)dudv \\ \frac{\partial^2 R_Y(t_1,t_2)}{\partial t_1 \partial t_2} &= R_X(t_1,t_2) \\ R_{XY}(t_1,t_2) &= \int_0^{t_2} R_X(t_1,u)du \\ \frac{\partial R_{XY}(t_1,t_2)}{\partial t_2} &= R_X(t_1,t_2) \end{aligned}$$

$$\therefore \quad E\left[\left(\frac{d}{dt}Y(t) - X(t)\right)^2\right] = R_X(t,t) - 2R_X(t,t) + R_X(t,t) = 0$$

9.89

$$\begin{aligned} R_{XX'}(t_1,t_2) &= \frac{\partial}{\partial t_2}R_X(t_1,t_2) \\ &= \frac{\partial}{\partial t_2}R_X(t_1-t_2) \\ &= -\frac{dR_X(\tau)}{d\tau}, \qquad \tau = t_1-t_2 \end{aligned}$$

$$R_X(\tau) \le R_X(0), \qquad R_X(0) \text{ is the max of } R_X(\tau)$$

$$\therefore \left.\frac{dR_X(\tau)}{d\tau}\right|_{\tau=0} = 0$$

$$\text{i.e.,} \qquad R_{XX'}(t,t) = 0$$

9.90 If $m_Z(t) = 0 \Rightarrow m_X(t) = 0$.
Proceeding as in Ex. 6.41:

$$R_{ZX}(t_1,t_2) = \int_0^{t_2} e^{-\alpha(t_2-\tau)}R_Z(t_1,\tau)d\tau = \int_0^{t_2} e^{-\alpha(t_2-\tau)}\sigma^2 e^{-\beta|t_1-\tau|}d\tau$$

We note that:

$$e^{-\beta|t_1-\tau|} = \begin{cases} e^{-\beta(t_1-\tau)} & \text{for } t_1 \ge \tau \\ e^{\beta(t_1-\tau)} & \text{for } t_1 \le \tau \end{cases}$$

We now suppose that $t_1 \le t_2$ so the above integral becomes:

$$\begin{aligned} R_{ZX}(t_1,t_2) &= \sigma^2 e^{-\alpha t_2}\left[\int_0^{t_1} e^{\alpha\tau}e^{-\beta t_1}e^{\beta\tau}d\tau + \int_{t_1}^{t_2} e^{\alpha\tau}e^{\beta t_1}e^{-\beta\tau}d\tau\right] \\ &= \sigma^2 e^{-\alpha t_2}\left[\frac{e^{-\beta t_1}(e^{(\alpha+\beta)t_1}-1)}{\alpha+\beta} + \frac{e^{\beta t_1}(e^{(\alpha-\beta)t_2}-e^{(\alpha-\beta)t_1})}{\alpha-\beta}\right] \end{aligned}$$

The autocorrelation of $X(t)$ is then

$$\begin{aligned} R_X(t_1,t_2) &= \int_0^{t_1} e^{-\alpha(t_1-\tau)}R_{ZX}(\tau,t_2)d\tau \\ &= \sigma^2 e^{-\alpha t_1}e^{-\alpha t_2}\left[\int_0^{t_1} e^{\alpha\tau}\frac{e^{-\beta\tau}(e^{(\alpha+\beta)\tau}-1)}{\alpha+\beta}d\tau\right. \\ &\quad \left. + \int_0^{t_1} e^{\alpha\tau}\frac{e^{\beta\tau}(e^{(\alpha-\beta)t_2}-e^{(\alpha-\beta)\tau})}{\alpha-\beta}d\tau\right] \\ &= \sigma^2 e^{-\alpha t_1}e^{-\alpha t_2}\left\{\frac{1}{\alpha+\beta}\left[\frac{1}{2\alpha}(e^{2\alpha t_1}-1) - \frac{1}{\alpha-\beta}(e^{(\alpha-\beta)t_1}-1)\right]\right. \\ &\quad \left. + + \frac{1}{\alpha-\beta}\left[\frac{e^{(\alpha-\beta)t_2}}{\alpha+\beta}(e^{(\alpha+\beta)t_1}-1) - \frac{1}{2\alpha}(e^{2\alpha t_1}-1)\right]\right\} \\ &= \frac{\sigma^2}{\alpha^2-\beta^2}\left[e^{-\beta(t_1-t_2)} - \frac{\beta}{\alpha}e^{-\alpha(t_2-t_1)} + \frac{\beta+\alpha}{\alpha}e^{-\alpha t_1-\alpha t_2}\right. \\ &\quad \left. -e^{-\alpha t_2-\beta t_1} - e^{-\beta t_2-\alpha t_1}\right] \end{aligned}$$

Letting $t_2 = t_1+\tau$, we see that as $t_1 \to \infty$ transient effects die out and the autocorrelation becomes

$$R_Z(t_1,t_2) \to \frac{\sigma^2}{\alpha-\beta^2}\left[e^{-\beta(t_2-t_1)} - \frac{\beta}{\alpha}e^{-\alpha(t_2-t_1)}\right]$$

9.8 Time Averages of Random Processes and Ergodic Theorems

9.91

6.78 $\mathcal{E}[X(t)] = \mathcal{E}[A] = 0 \qquad \mathcal{E}[X(t_1)X(t_2)] = \mathcal{E}[A^2] = 1$

$$\begin{aligned} VAR[< X(t) >_T] &= \frac{1}{2T}\int_{-2T} 2T\left(1 - \frac{|u|}{2T}\right) C_X(u)du \\ &= \frac{2}{2T}\int_0^{2T}\left(1 - \frac{u}{2T}\right) du \\ &= 1 \end{aligned}$$

$\Rightarrow$ process is not mean-ergodic.

9.92)

a) First we have to check for WSS

$E[X_n] = \frac{21}{6}$, $C_X(n, n+k) = 2.9167$

Therefore X_n is WSS

we have to find:

$$VAR[\langle X_n \rangle_T] = \frac{1}{2T+1} \sum_{k=-2T}^{2T} \left(1 - \frac{|k|}{2T+1}\right) C_X(k)$$

$$= \frac{2.9167}{2T+1} \sum_{k=-2T}^{2T} \left(1 - \frac{|k|}{2T+1}\right) = 2.9167\left(1 - \frac{\sum |k|}{(2T+1)^2}\right)$$

it can be seen that $\lim_{T \to \infty} [\langle X_n \rangle_T] = \lim_{T \to \infty} 2.9167\left(1 - \frac{2T(2T+1)}{(2T+1)^2}\right)$

$= 0$

Therefore X_n is mean ergodic.

b) $E[X_n] = 0$, $C_X(n, n+k) = \begin{cases} 1, & k \text{ even} \\ -1, & k \text{ odd} \end{cases} = (-1)^k$

Therefore X_n is WSS

$$VAR[\langle X_n \rangle_T] = \frac{1}{2T+1} \sum_{k=-2T}^{2T} \left(1 - \frac{|k|}{2T+1}\right) C_X(k)$$

$$= \frac{1}{2T+1}\left(1 - \sum_{k=-2T}^{2T} \frac{|k|(-1)^k}{2T+1}\right) = \frac{1}{2T+1}\left(1 - \frac{2\sum_{1}^{2T} k(-1)^k}{2T+1}\right)$$

$= \frac{1}{2T+1}\left(1 - \frac{2T}{2T+1}\right)$, which $\to 0$ if $T \to \infty$ then X_n is mean ergodic

P9.92)

c) $X_n = s^n \quad n \geq 0 \quad s \sim U(0,1)$

$$E[X_n] = E[s^n] = \int_0^1 s^n ds = \frac{1}{n+1}$$

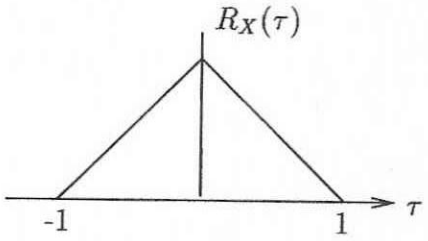
$$C_X(n, n+k) = E[X_n X_{n+k}] - E[X_n]E[X_{n+k}] = E[s^{2n+k}] - \frac{1}{(n+1)} \times \frac{1}{(n+k+1)}$$

$$= \frac{1}{2n+k+1} - \frac{1}{(n+1)(n+k+1)}$$

which shows X_n is not wss, then we can't check for mean ergodicity using Eqn. (9.108)

9.93)

6.79 $R_X(\tau) = A(1 - |\tau|), \quad |\tau| \leq 1$



$$\begin{aligned} VAR[< X(t) >_T] &= \frac{1}{2T}\int_{-2T}^{2T}\left(1 - \frac{|u|}{2T}\right) R_X(u)du \\ &< \frac{1}{2T}\int_{-2T}^{2T} R_X(u)du \\ &= \frac{1}{2T}\int_{-2T}^{2T} A(1-|u|)du \\ &= \frac{1}{2T}\left(\frac{A}{2}\right) \quad \text{for } T > 1 \\ &\to 0 \quad \text{as } T \to \infty \end{aligned}$$

$\Rightarrow X(t)$ is mean-ergodic.

9.94

6.80 a) $\mathcal{E}[X(t)] = \mathcal{E}[A]\cos 2\pi ft = m\cos 2\pi ft$

$$\begin{aligned} < X(t) >_T &= \frac{1}{2T}\int_{-T}^{T} X(t)dt = \frac{1}{2T}\int_{-T}^{T} A\cos 2\pi ftdt \\ &= \frac{1}{2T}\frac{2A\sin 2\pi fT}{2\pi f} \to 0 \text{ as } T \to \infty \end{aligned}$$

b)
$$\begin{aligned} R_X(t,t+\tau) &= \mathcal{E}[A^2]\cos 2\pi ft\cos 2\pi f(t+\tau) \\ < X(t)X(t+\tau) >_T &= \frac{1}{2T}\int_{-T}^{T} A^2\cos 2\pi f(t+\tau)\cos 2\pi ftdt \\ &= \frac{1}{2T}\int_{-T}^{T}\frac{A^2}{2}\left[\cos 2\pi f\tau + \frac{A^2}{2}\cos 2\pi f(2t+\tau)\right]dt \\ &\to \frac{A^2}{2}\cos 2\pi f\tau \end{aligned}$$

9.95

6.81 a)
$$\begin{aligned} \mathcal{E}[X(t)] &= \mathcal{E}[A]\mathcal{E}[\cos(2\pi ft+\Theta)] = 0 = m_X(t) \\ < X(t) >_T &= \frac{1}{2T}\int_{-T}^{T} A\cos(2\pi ft+\Theta) = \frac{A\sin(2\pi fT+\Theta)}{2\pi fT} \to 0 = m_X(t) \end{aligned}$$

b)
$$\begin{aligned} \mathcal{E}[X(t)X(t+\tau)] &= \mathcal{E}[A^2]\mathcal{E}[\cos(2\pi ft+\Theta)\cos(2\pi f(t+\tau)+\Theta)] \\ &= \frac{\mathcal{E}[A^2]}{2}\cdot 2\pi f\tau = R_X(\tau) \\ < X(t)X(t+\tau) >_T &= \frac{1}{2T}\int_{-T}^{T} A^2\cos(2\pi f\tau+\Theta)\cos(2\pi f(t+\tau)+\Theta)dt \\ &= \frac{1}{2T}\int_{-T}^{T} A^2[\cos 2\pi f\tau + \cos(2\pi f(2t+\tau)+\Theta)]dt \\ &\to \frac{A^2}{2}\cos 2\pi f\tau \neq R_X(\tau) \end{aligned}$$

9.96

$$\begin{aligned} VAR[< X(t) >_T] &= \frac{1}{T}\int_0^{2T}\left(1-\frac{u}{2T}\right)e^{-2\alpha u}du \\ &= \frac{1}{T}\int_0^{2T} e^{-2\alpha u}du - \frac{1}{2T^2}\int_0^{2T} ue^{-2\alpha u}du \\ &= \frac{4\alpha T + e^{-4\alpha T} - 1}{8\alpha^2T^2} \quad \to \ 0 \end{aligned}$$

9.97

$$\begin{aligned} VAR[< X_n >_T] &= \frac{1}{2T+1}\sum_{k=-2T}^{2T}\left(1-\frac{|k|}{2T+1}\right)C_X(k) < \frac{1}{2T+1}\sum_{k=-2T}^{2T}\left(\frac{1}{2}\right)^{|k|} < \frac{2}{2T+1} \\ &\to 0 \text{ as } T \to \infty \\ &\Rightarrow X_n \text{ is mean-ergodic} \end{aligned}$$

9.98

a) $Y_n = \frac{1}{2}(X_n + X_{n-1})$, $X_0 = 0$ & X_n ~ Bernoulli(p)

all the computations are for $n \geq 2$

$$E[X_n] = p \quad , \quad R_X(k) = E[X_n X_{n+k}] = \begin{cases} p^2 & k \neq 0 \\ p & k = 0 \end{cases} \qquad C_Y(k) = \begin{cases} p(1-p) & k = 0 \\ 0 & k \neq 0 \end{cases}$$

$$R_Y(k) = E[Y_n Y_{n+k}] = \frac{1}{4}\left[R_X(k+1) + 2R_X(k) + R_X(k-1)\right]$$

$$E[Y_n] = p \quad , \quad C_Y(k) = R_Y(k) - p^2$$

Y_n is WSS

$$C_Y(k) = \begin{cases} \frac{1}{4}[2p^2 + 2p] - p^2 = \frac{1}{2}p - \frac{1}{2}p^2 & k = 0 \\ \frac{1}{4}(p + 3p^2) - p^2 = \frac{1}{4}p - \frac{1}{4}p^2 & k = \pm 1 \\ \frac{1}{4}[4p^2] - p^2 = 0 & |k| > 1 \end{cases}$$

It can be easily shown that the following limit is zero

$$\lim_{T \to \infty} VAR[\langle Y_n \rangle_T] = \lim_{T \to \infty} \frac{1}{2T+1} \sum_{k=-2T}^{2T} \left(1 - \frac{|k|}{2T+1}\right) C_Y(k) = 0$$

Then Y_n is mean ergodic.

(b) Since $E[Z_n] = m_X \frac{1-(\frac{3}{4})^{n-1}}{1-\frac{3}{4}}$ depends on n, Z_n is not WSS.

However if we suppose the process was started at time $-\infty$, then

$$E[Z_n] = m_X$$

and $$Cov(Z_n, Z_{n+k}) = \frac{(\frac{3}{4})^{|k|}}{1-(\frac{3}{4})^2}\sigma_X^2 \quad \text{(see problem 9.32)}.$$

$$= \beta\, \alpha^{|k|}$$

We then have

$$VAR[\langle X_n \rangle_T] < \frac{2}{2T+1}\beta \sum_{k=0}^{2T} \alpha^{|k|} = \frac{2\beta}{2T+1}\frac{1}{1-\alpha} \to 0 \quad \text{as } T \to \infty$$

∴ the steady state process is mean-ergodic

9.99 ~~6.84~~

(a)

$$\frac{1}{2T}\int_{-2T}^{2T}\left(1-\frac{|u|}{2T}\right)C_X(u)du < \frac{1}{2T}\int_{-2T}^{2T}|C_X(u)|du$$

$\therefore$ If $\int_{-\infty}^{\infty}|C_X(u)|du = M < \infty$ then

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-2T}^{2T}\left(1-\frac{|u|}{2T}\right)C_X(u)du < \lim_{T\to\infty}\frac{1}{2T}\int_{-2T}^{2T}|C_X(u)|du = 0$$

$\Rightarrow X(t)$ is mean-ergodic.

(b)

$$\lim_{T\to\infty}\frac{1}{2T+1}\sum_{k=-2T}^{2T}\left(1-\frac{|k|}{2T+1}\right)C_X(k) < \lim_{T\to\infty}\frac{1}{2T+1}\sum_{k=-2T}^{2T}|C_X(k)| <$$

$$< \lim_{T\to\infty}\frac{M}{2T+1} = 0.$$

$\Rightarrow X_n$ is mean ergodic

9.100

6.85 In order for $< X^2(t) >_T$ to be a valid estimate for $\mathcal{E}[X^2(t)]$, the process $Y(t) = X^2(t)$ must be mean-ergodic. Since $\mathcal{E}[X^2(t)] = C_X(0)$, a constant, $X^2(t)$ is mean-ergodic iff $C_{X^2}(t_1, t_2)$ is such that

$$\lim_{T\to\infty} \frac{1}{4T^2}\int_{-T}^{T}\int_{-T}^{T} C_{X^2}(t_1,t_2)dt_1dt_2$$
$$= \lim_{T\to\infty} \frac{1}{4T^2}\int_{-T}^{T}\int_{-T}^{T}[\mathcal{E}[X^2(t_1)X^2(t_2)] - C_X^2(0)]dt_1dt_2 = 0$$

Note that, in general, $X(t)$ WSS does not necessarily imply that $X^2(t)$ is WSS.

(b)

$X(t) = A\cos(2\pi t + \Theta)$ Θ uniform $(0, 2\pi)$

$X^2(t) = A^2\cos^2(2\pi t + \Theta) = \frac{A^2}{2} + \frac{A^2}{2}\cos(4\pi t + 2\Theta)$

$E[X^2(t)] = \frac{A^2}{2}$

$C_{X^2}(t_1,t_2) = E\left[\left(X^2(t_1) - \frac{A^2}{2}\right)\left(X^2(t_2) - \frac{A^2}{2}\right)\right]$

$= E\left[\frac{A^2}{2}\cos(4\pi t_1 + 2\Theta)\frac{A^2}{2}\cos(4\pi t_2 + 2\Theta)\right]$

$= \frac{A^4}{8}E\left[\cos 4\pi(t_2 - t_1) + \cos(4\pi(t_1+t_2) + 4\Theta)\right]$

$= \frac{A^4}{8}\cos 4\pi(t_2 - t_1)$.

Applying theorem for mean ergodicity:

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-2T}^{2T}\left(1 - \frac{|\tau|}{2T}\right)\frac{A^4}{8}\cos 4\pi\tau\, d\tau$$

$$< \frac{1}{2T}\frac{A^4}{8}\int_{-2T}^{2T}\cos 4\pi\tau\, d\tau$$

$$\frac{1}{4\pi}\sin 4\pi\tau\Big|_{-2T}^{2T}$$

$$\frac{2}{4\pi}\sin 8\pi T$$

$$< \frac{A^4}{32\pi}\frac{1}{T} \to 0 \Rightarrow X^2(t) \text{ is mean ergodic}$$

$$\langle X^2(t)\rangle_T = \frac{1}{2T}\int_{-T}^{T}\left(\frac{A}{2} + \frac{A}{2}\cos(4\pi t + 2\Theta)\right)dt \to \frac{A^2}{2}.$$

9.101a In order for $< X(t)X(t+\tau) >_T$ to be a valid estimate for $R_X(\tau)$, $Y(t) = X(t)X(t+\tau)$ must be mean-ergodic.

Note that

$$\mathcal{E}[< X(t)X(t+\tau) >_T] = \frac{1}{2T}\int_{-T}^{T} \mathcal{E}[X(t)X(t+\tau)]dt = R_X(\tau)$$

does not depend on t. Thus $X(t)X(t+\tau)$ is mean ergodic iff $C_{X(t)X(t+\tau)}(t_1, t_2)$ is such that

$$\lim_{T\to\infty} \frac{1}{4T^2}\int_{-T}^{T}\int_{-}^{T} C_{X(t)X(t+\tau)}(t_1,t_2)dt_1dt_2$$
$$= \lim_{T\to\infty} \frac{1}{4T^2}\int_{-T}^{T}\int_{-T}^{T}[\mathcal{E}[X(t_1)X(t_1+\tau)X(t_2)X(t_2+\tau)] - R_X^2(\tau)]dt_1dt_2$$
$$= 0$$

(b) $X(t) = A\cos(2\pi t + \Theta)$ Θ uniform in $(0, 2\pi)$

$$X(t)X(t+\tau) = A^2\cos(2\pi t+\Theta)\cos(2\pi(t+\tau)+\Theta)$$
$$= \frac{A^2}{2}\left[\cos 2\pi\tau + \cos(2\pi(2t+\tau)+2\Theta)\right]$$

$$E[X(t)X(t+\tau)] = \frac{A^2}{2}\cos 2\pi\tau$$

$$C_{X(t)X(t+\tau)}(t_1,t_2) = E\left[\frac{A^2}{2}\cos(2\pi(2t_1+\tau)+2\Theta)\frac{A^2}{2}\cos(2\pi(2t_2+\tau)+2\Theta)\right]$$
$$= \frac{A^4}{8}E\left[\cos 4\pi(t_2-t_1) + \cos(2\pi(2t_1+2t_2+2\tau)+4\Theta)\right]$$
$$= \frac{A^4}{8}\cos 4\pi(t_2-t_1)$$

Applying the theorem on ergodicity:

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-2T}^{2T}\left(1-\frac{|\alpha|}{2T}\right)\frac{A^4}{8}\cos 4\pi\tau\, d\tau$$
$$< \frac{1}{2T}\cdot\frac{A^4}{8}\cdot\frac{2}{4\pi}\sin 8\pi T$$
$$< \frac{A^4}{32\pi}\cdot\frac{1}{T} \to 0 \Rightarrow X(t)X(t+\tau) \text{ is mean ergodic}$$

$$\langle X(t)X(t+\tau)\rangle_T = \frac{1}{2T}\int_{-T}^{T}\frac{A^2}{2}\cos 2\pi\tau + \frac{A^2}{2}\cos(2\pi(2t+\tau)+2\Theta)\,dt$$
$$\to \frac{A^2}{2}\cos 2\pi\tau \quad \checkmark$$

9.101

a) $< Y(T) >_T = \frac{1}{2T}\int_{-T}^{T} Y(t)dt = \frac{1}{2T}$ [total time $X(t) \in (a, b]$ during $t \in [-T, T]$]

b)
$$\begin{aligned}\mathcal{E}[< Y(t) >_T] &= \frac{1}{2T}\int_{-T}^{T} \mathcal{E}[Y(t)]dt \\ &= \frac{1}{2T}\int_{-T}^{T} P[a < X(t) \leq b]dt\end{aligned}$$

If $P[a < X(t) \leq b]$ does not depend on t, then

$$\mathcal{E}[< Y(t) >_T] = P[a < X(t) \leq b] \triangleq P[a < X \leq b]$$

(c) $< Y(t) >_T \to P[a < X \leq b]$ if $Y(t)$ is mean ergodic. That is, if

$$\begin{aligned}&\lim_{T\to\infty} \frac{1}{4T^2}\int_{-T}^{T}\int_{T}^{T} C_Y(t_1, t_2)dt_1dt_2 \\ &= \lim_{T\to\infty} \frac{1}{4T^2}\int_{-T}^{T}\int_{-T}^{T} \{\mathcal{E}[Y(t_1)Y(t_2)] - P[a < X \leq b]^2\}dt_1dt_2 \\ &= \lim_{T\to\infty} \frac{1}{4T^2}\int_{-T}^{T}\int_{-T}^{T} \{P[X(t_1) \in (a, b], X(t_2) \in (a, b]] - P^2[a < X \leq b]\}dt_1dt_2 \\ &= 0\end{aligned}$$

d) Let $a = -\infty$.

e) For the random telegraph signal let $a = -\infty$, so we seek $P[X(t) \leq 0]$. Assume $X(t)$ wp

$Y(t) = I_{\{X(t)<0\}}$

$E[Y(t)] = E[I_{\{X(t)<0\}}] = P[X(t)<0] = \frac{1}{2}$

$C_Y(t_1, t_2) = E[I_{\{X(t_1)<0\}} I_{\{X(t_2)<0\}}] - \left(\frac{1}{2}\right)^2$

$= P[\text{even \# of transitions in } (t_1, t_2) \mid X(t_1)<0]\frac{1}{2} - \frac{1}{4}$

$= \frac{1}{2}\left(1 + e^{-2\alpha|t_2 - t_1|}\right)\frac{1}{2} - \frac{1}{4} = \frac{1}{2}e^{-2\alpha|t_2 - t_1|}$

Applying theorem on mean ergodicity

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T}\left(1 - \frac{|\tau|}{T}\right)\frac{1}{2}e^{-2\alpha|\tau|}d\tau$$

$$< \lim_{T\to\infty} \frac{2}{4T}\int_{0}^{T} e^{-2\alpha\tau}d\tau$$

$$= \lim_{T\to\infty} \frac{1}{2\alpha T}\left(1 - e^{-2\alpha T}\right) \to 0$$

$\Rightarrow$ mean ergodic

9.103

6.88 **a)** $< Y_n >_T = \frac{1}{2T+1}\sum_{n=-T}^{T} Y_n = \frac{1}{2T+1}$ [# occurrences of $\{a < X_n \leq b\}$ during $t \in \{-T, T\}$]

b)
$$\mathcal{E}[< Y_n >_T] = \frac{1}{2T+1}\sum_{n=-T}^{T}\mathcal{E}[Y_n] = \frac{1}{2T+1}\sum_{n=-T}^{T} P[a < X_n \leq b]$$
$$= P[a < X \leq b] \quad \text{if } P[a < X_n \leq b] \text{ does not depend on } n$$

c) Y_n is mean-ergodic iff

$$\lim_{T\to\infty}\frac{1}{(2T+1)^2}\sum_{k=-T}^{T}\sum_{j=-T}^{T}\{P[X_k \in (a,b], X_j \in (a,b]] - P^2[a < X \leq b]\} = 0$$

d) $a = -\infty$, $b = x$.

9.104

6.89 **a)** Here we suppose that we observe X_n only for $n \leq 1$

$$Z_n = u(a - X_n)$$
$$\frac{1}{n}\sum_{k=1}^{n} Z_n = \frac{1}{n}\underbrace{\sum_{k=1}^{n} u(a - X_n)}_{\substack{\text{counting process for} \\ \text{event } \{X_n \leq a\}}}$$

b) If Z_n is mean-ergodic, then

$$\frac{1}{n}\sum_{k=1}^{n} Z_n \to \mathcal{E}[Z_n] = \mathcal{E}[u(a - X_n)] = P[X_n \leq a] = F_X(a)$$

9.105

$$C_X(k) = \frac{\delta^2}{2}\left\{ |k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H} \right\}$$

$$VAR[\langle X_n \rangle_T] = \frac{1}{(2T+1)^2}\left\{ (2T+1)C_X(0) + 2\times 2T C_X(1) + 2\times(2T-1)C_X(2) + \cdots + 2C_X(2T) \right\}$$

if you replace $C_X(k)$s in the VAR equation you can see:

$$(2T+1)C_X(0) = (2T+1)\frac{\delta^2}{2}\left\{ 1^{2H} - 2\times|0|^{2H} + 1^{2H} \right\}$$

$$2\times 2T \quad C_X(1) = 2(2T)\frac{\delta^2}{2}\left\{ 2^{2H} - 2\times 1^{2H} + 0^{2H} \right\}$$

$$2\times(2T-1) \quad C_X(2) = 2(2T-1)\frac{\delta^2}{2}\left\{ 3^{2H} - 2\times 2^{2H} + 1^{2H} \right\}$$

$$\vdots$$

$$2\times 2 \quad C_X(2T-1) = 2\times 2\times\frac{\delta^2}{2}\left\{ (2T)^{2H} - 2\times(2T-1)^{2H} + (2T-2)^{2H} \right\}$$

$$2\times 1 \quad C_X(2T) = 2\times 1\times\frac{\delta^2}{2}\left\{ (2T+1)^{2H} - 2\times(2T)^{2H} + (2T-1)^{2H} \right\}$$

$$VAR[\langle X_n \rangle_T] = \frac{1}{(2T+1)^2}\left((2T+1)^{2H} \right) \times 2\times 1\times\frac{\delta^2}{2} = \delta^2(2T+1)^{2H-2}$$

9.106

```
T=1:5:100;
%P.9.106
H2=2*0.5;Cx1=0.5*((T+1).^H2-2*T.^H2+(T-1).^H2);
H2=2*0.6;Cx2=0.5*((T+1).^H2-2*T.^H2+(T-1).^H2);
H2=2*0.75;Cx3=0.5*((T+1).^H2-2*T.^H2+(T-1).^H2);
H2=2*0.99;Cx4=0.5*((T+1).^H2-2*T.^H2+(T-1).^H2);
plot(T, Cx1, '-', T, Cx2, '-*', T,Cx3, '-^', T,Cx4, '--');
legend('H=0.5', 'H=0.6', 'H=0.75', 'H=0.99');
title('Problem 9.106');
%as it can be seen long range dependence increases with H
```

9.107

(a)

```
T=1:5:100;
%P.9.107a
H=0.5;var1=(2*T+1).^(2*H-2);
H=0.6;var2=(2*T+1).^(2*H-2);
H=0.75;var3=(2*T+1).^(2*H-2);
H=0.99;var4=(2*T+1).^(2*H-2);
figure(2);
plot(T, var1, '-', T, var2, '-*', T,var3, '-^', T,var4, '--');
legend('H=0.5', 'H=0.6', 'H=0.75', 'H=0.99');
title('Problem 9.107a');
```

(b)

```
T=1:5:100;
%P.9.107b
H=0.5;var1=(2*T+1).^(2*H-1);
H=0.6;var2=(2*T+1).^(2*H-1);
H=0.75;var3=(2*T+1).^(2*H-1);
H=0.99;var4=(2*T+1).^(2*H-1);
figure(3);
plot(T, var1, '-', T, var2, '-*', T,var3, '-^', T,var4, '--');
legend('H=0.5', 'H=0.6', 'H=0.75', 'H=0.99');
title('Problem 9.107b');
```

9.108

(a)

```
T=1:5:100;
%P.9.106
H=0.5;var1=(2*T+1).^(2*H-2);
H=0.6;var2=(2*T+1).^(2*H-2);
H=0.75;var3=(2*T+1).^(2*H-2);
H=0.99;var4=(2*T+1).^(2*H-2);
figure(4);
loglog(T, var1, '-', T, var2, '-*', T,var3, '-^', T,var4, '--');
legend('H=0.5', 'H=0.6', 'H=0.75', 'H=0.99');
title('Problem 9.106');
%answer is H=0.75;
```

9.109

$$C_X(k) = \sigma^2 H(2H-1) k^{2H-2}$$

$\sum_{k=-\infty}^{+\infty} |C_X(k)| < \infty$ is the sufficient condition.

$$\sum_{k=-\infty}^{+\infty} |C_X(k)| = \sigma^2 H(2H-1) \sum_{k=-\infty}^{+\infty} |k|^{2H-2} = 2\sigma^2 H(2H-1) \sum_{k=1}^{\infty} k^{2H-2}$$

$$= 2\sigma^2 H(2H-1) \sum_{k=1}^{\infty} k^{-\alpha} \qquad \alpha = 2-2H, \ \frac{1}{2} < H < 1 \Rightarrow 0 < \alpha < 1$$

This sum does not converge and is not bounded.

Although the sufficient condition is not met, the process is still mean ergodic as discussed in Example 9.50

*9.9 Fourier Series and Karhunen-Loeve Expansion

9.110

a) $X(t) = Xe^{j\omega t}$

$$R_X(t_1, t_2) = E[X(t_1)X^*(t_2)] = E[Xe^{j\omega t_1} X^* e^{-j\omega t_2}]$$

$$= E[XX^*]e^{j\omega(t_1 - t_2)} = E[|X|^2]\, e^{j\omega(t_1 - t_2)}$$

b) $E[X(t)] = E[X]e^{j\omega t}$

$R_X(\tau) = E[|X|^2]e^{j\omega\tau}$

if $E[X] = 0$, then $X(t)$ is a WSS random process.

9.111) $X(t) = X_1 e^{j\omega_1 t} + X_2 e^{j\omega_2 t}$, $\omega_1 \neq \omega_2$

a) $E[X(t_1)X^*(t_2)] = E\left[(X_1 e^{j\omega_1 t_1} + X_2 e^{j\omega_2 t_1})(X_1^* e^{-j\omega_1 t_2} + X_2^* e^{-j\omega_2 t_2})\right]$

$$= E[|X_1|^2] e^{j\omega_1(t_1-t_2)} + E[|X_2|^2] e^{j\omega_2(t_1-t_2)}$$

$$+ E[X_1 X_2^*] e^{j(\omega_1 t_1 - \omega_2 t_2)} + E[X_2 X_1^*] e^{j(\omega_2 t_1 - \omega_1 t_2)}$$

$$E[X(t)] = E[X_1] e^{j\omega_1 t} + E[X_2] e^{j\omega_2 t}$$

$$C_X(t_1,t_2) = E[X(t_1)X_2^*(t)] - E[X(t_1)]\,E[X^*(t_2)]$$

$$E[X(t_1)]\,E[X^*(t_2)] = E[X_1]\,E[X_1^*] e^{j\omega_1(t_1-t_2)} + E[X_2]E[X_2^*] e^{j\omega_2(t_1-t_2)}$$

$$+ E[X_1]\,E[X_2^*] e^{j(\omega_1 t_1 - \omega_2 t_2)} + E[X_2]\,E[X_1^*] e^{j(\omega_2 t_1 - \omega_1 t_2)}$$

Therefore:

$$C_X(t_1,t_2) = VAR[X_1] e^{j\omega_1(t_1-t_2)} + VAR[X_2] e^{j\omega_2(t_1-t_2)}$$

$$+ cov(X_1,X_2) e^{j(\omega_1 t_1 - \omega_2 t_2)} + cov(X_2,X_1) e^{j(\omega_2 t_1 - \omega_1 t_2)}$$

b) $E[X(t)] = E[X_1] e^{j\omega_1 t} + E[X_2] e^{j\omega_2 t}$, $\omega_1 \neq \omega_2$

first condition: $E[X_1] = E[X_2] = 0$

Since $\omega_1 \neq \omega_2$

another condition is: X_1 & X_2 should be uncorrelated.

so that $C_X(t_1,t_2)$ could be function of $t_1 - t_2$ only.

c) $X(t) = \frac{(U-jV)}{2}e^{jwt} + \frac{(U+jV)}{2}e^{-jwt}$, $w_1 = -w_2 = w$

$= \frac{U}{2}(e^{jwt} + e^{-jwt}) - j\frac{V}{2}(e^{jwt} - e^{-jwt})$

$= U\cos wt + V\sin wt$, $X(t)$ is real.

$R_X(t_1,t_2) = E[X(t_1)X(t_2)] = E[(U\cos wt_1 + V\sin wt_1)(U\cos wt_2 + V\sin wt_2)]$

$= E[U^2]\cos wt_1\cos wt_2 + E[V^2]\sin wt_1\sin wt_2 + E[UV]\sin(wt_1 + wt_2)$

d) $E[X(t)] = E[U]\cos wt + E[V]\sin wt$

first condition $E[U] = E[V] = 0$, and therefore $E[X(t)] = 0$ $\forall t$

Also:

$C_X(t_1,t_2) = R_X(t_1,t_2) - E[X(t_1)]E[X(t_2)] = R_X(t_1,t_2)$

second condition: $E[U^2] = E[V^2]$, and U & V should be orthogonal

e) $X(t) = U\cos wt + V\sin wt$

$X(t_1)$ is a linear combination of two jointly Gaussian Random Variables.

Then $X(t_1)$ is also a Gaussian Random Variable.

(chapter 6)

Similarly, it can be easily shown that $X(t_1)$ & $X(t_2)$ ... & $X(t_k)$ are jointly Gaussian Random variables.

Therefore $X(t)$ is a Gaussian Random process.

9.112

6.90 a The correlation between Fourier coefficients is:

$$\begin{aligned} E[X_k X_m^*] &= E\left[\frac{1}{T}\int_0^T X(t')e^{-j2\pi kt'/T}dt'\frac{1}{T}\int_0^T X(t'')e^{j2\pi mt''/T}dt''\right] \\ &= \frac{1}{T^2}\int_0^T\int_0^T R_X(t'-t'')e^{-j2\pi kt'/T}e^{j2\pi mt''/T}dt'dt'' \end{aligned}$$

This is Eqn.9.118..

b) Now suppose $X(t)$ is M.S. periodic:

$$\begin{aligned} E[X_k X_m^*] &= \frac{1}{T^2}\int_0^T e^{j2\pi nt''/T}dt''\int_0^T R_X(t't-'')e^{-j2\pi k't/T}dt' \\ &= \frac{1}{T^2}\int_0^T e^{j2\pi mt''/T}dt''\int_{-t''}^{T-t''} R_X(u)e^{-j2\pi k(u+t'')}du \\ &= \frac{1}{T}\int_0^T e^{j2\pi(m-k)t''/T}dt''\frac{1}{T}\int_{-t''}^{T-t''} R_X(u)e^{-j2\pi ku}du \end{aligned}$$

If $X(t)$ is M.S. periodic then $R_X(u)$ is periodic and the inner integral is a_k, thus

$$\begin{aligned} E[X_k X_m^*] &= a_k\frac{1}{T}\int_0^T e^{j2\pi(m-k)t''/T}dt'' \\ &= a_k\delta_k m \qquad \surd \end{aligned}$$

9.113 $X(t)$ WSS Gaussian RP, $R_X(\tau) = e^{-|\tau|}$

$X(t)$ is not mean square periodic, but we expand it in the interval $[0, T]$

$$X(t) = \sum_{k=-\infty}^{\infty} X_k e^{j2\pi kt/T} \qquad 0 \le t \le T$$

where

$$X_k = \frac{1}{T}\int_0^T X(t')e^{-j2\pi kt'/T}\,dt'$$

X_k is defined by a linear transformation of a jointly Gaussian RP so it is also a Gaussian RV

mean $E[X_k] = \frac{1}{T}\int_0^T E[X(t')]\,e^{-j2\pi kt'/T}\,dt'$

$= \frac{1}{T}\int_0^T \underbrace{m_X(t)}_{=0}\,e^{-j2\pi kt'/T}\,dt'$ assume zero mean WSS

$= 0$

By Eq. 9.118

$$E[X_k X_k^*] = \frac{1}{T^2}\int_0^T\int_0^T e^{-|t-u|}\,e^{-j2\pi\frac{kt}{T}}\,e^{j2\pi\frac{ku}{T}}\,dt\,du$$

Variance →

$$= \frac{1}{T^2}\int_{-T}^{T} e^{-|\tau|}\,e^{-j\frac{2\pi k}{T}\tau}\,(T-|\tau|)\,d\tau$$

Note we can also write

$$X(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} A_k \cos\frac{2\pi kt}{T} + B_k \sin\frac{2\pi kt}{T}$$

where $A_k = X_k + X_{-k}$

$B_k = j(X_k - X_{-k})$

$$E[X_k X_k^*] = \frac{1}{T}\int_{-T}^{T} e^{-|\tau|}\cos\frac{2\pi k\tau}{T}\left(1-\frac{|\tau|}{T}\right)d\tau \qquad b=\frac{2\pi k}{T}$$

$$= \frac{2}{T}\int_0^T e^{-\tau}\cos b\tau\, d\tau - \frac{2}{T}\int_0^T \frac{\tau}{T} e^{-\tau}\cos b\tau\, d\tau$$

$$\int_0^T e^{-\tau}\cos b\tau d\tau = \frac{e^{-\tau}}{1+b^2}\left\{-\cos b\tau + b\sin b\tau\right\}_0^T = \frac{1-e^{-T}}{1+b^2}$$

$$\int_0^T \tau e^{-\tau}\cos b\tau d\tau = \left[\frac{\tau e^{-\tau}}{1+b^2}\left\{-\cos b\tau + b\sin b\tau\right\} - \frac{1}{1+b^2}\int e^{-\tau}(-\cos b\tau + b\sin b\tau)d\tau\right]_0^T$$

$$\int_0^T e^{-\tau}\sin b\tau d\tau = \frac{e^{-\tau}}{1+b^2}\left\{-\sin b\tau - b\cos b\tau\right\}_0^T = \frac{b}{1+b^2}(1-e^{-T})$$

$$\Rightarrow \int_0^T \tau e^{-\tau}\cos b\tau d\tau = \frac{-Te^{-T}}{1+b^2} + \frac{1-e^{-T}}{(1+b^2)^2} - \frac{b^2}{(1+b^2)^2}(1-e^{-T})$$

Finally $E[X_k X_k^*] = \frac{2}{T}\frac{1-e^{-T}}{1+b^2}$

$$-\frac{2}{T^2}\left\{\frac{-Te^{-T}(1+b^2)+(1-e^{-T})-b^2(1-e^{-T})}{(1+b^2)^2}\right.$$

Note that

$$E[X_k X_m^*] \neq 0$$

9.114

~~9.91~~ Assume $X(t)$ is a WSS mean square periodic process

$$K_X(t_1, t_2) = K_X(t_2 - t_1)$$

Eigenvalue equation:

$$\int_0^T K_X(t_1 - t_2)\phi_k(t_2)dt_2 = \lambda_k \phi_k(t_1)$$

Try

$$\phi_k(t) = \frac{1}{\sqrt{T}} e^{j\frac{2\pi}{T}kt}$$

$\{\phi_k(t)\}$ is orthonormal set and

$$\begin{aligned}
\int_0^T K_X(t_2 - t_1)\frac{1}{\sqrt{T}} e^{j\frac{2\pi}{T}kt_2} dt_2 &= \int_{-t_1}^{T-t_1} K_X(u)\frac{1}{\sqrt{T}} e^{j\frac{2\pi}{T}ku} e^{+j\frac{2\pi}{T}kt_1} du \\
&= \frac{1}{\sqrt{T}} e^{j\frac{2\pi}{T}kt_1} \int_0^T K_X(u) e^{j\frac{2\pi}{T}ku} du \\
&= \frac{1}{\sqrt{T}} e^{j\frac{2\pi}{T}kt_1} \cdot \lambda_k \\
&= \lambda_k \phi_k(t_1)
\end{aligned}$$

where the eigenvalue is given by

$$\lambda_k = \int_0^T K_X(u) e^{j\frac{2\pi}{T}ku} du \ .$$

Therefore the KL Expansion is:

$$X(t) = \sum_{k=-\infty} X_k \phi_k(t)$$

where

$$\begin{aligned}
X_k &= \int_0^T X(t)\phi_k^*(t)dt \\
&= \int_0^T X(t)\frac{1}{\sqrt{T}} e^{-j\frac{2\pi}{T}kt} dt
\end{aligned}$$

Thus

$$\frac{X_K}{\sqrt{T}} = \frac{1}{T}\int_0^T X(t) e^{-j\frac{2\pi}{T}kt} dt, \quad \text{the Fourier coefficients}$$

Therefore $K - L$ expansion of $X(t)$ yields the Fourier series.

9.115

6.92 For white Gaussian noise process

$$K_X(t_1, t_2) = \alpha\delta(t_1 - t_2)$$

Take any set of orthonormal functions $\{\phi_k(t)\}$

$$\begin{aligned}\int_0^T K_X(t_1, t_2)\phi_k(t_2)dt_2 &= \int_0^T \alpha\delta(t_1 - t_2)\phi_k(t_2)dt_2 \\ &= \alpha\phi_k(t_1)\end{aligned}$$

The eigenvalue equation is satisfied and $\lambda_k = \alpha$.

9.116

$$\begin{aligned}R_{XW}(t_1, t_2) &= 0 \\ K_W(t_1, t_2) &= \alpha\delta(t_1 - t_2) \\ \int_0^T K_X(t_1, t_2)\phi_n(t_2)dt_2 &= \lambda_n\phi_n(t_1) \\ K_Y(t_1, t_2) &= E[(X(t_1) + W(t_1))(X(t_2) + W(t_2))] \\ &= K_X(t_1, t_2) + K_W(t_1, t_2) \\ \int_0^T K_Y(t_1, t_2)\phi_n(t_2)dt_2 &= \int_0^T [K_X(t_1, t_2) + \alpha\delta(t_1 - t_2)]\phi_n(t_2)dt_2 \\ &= \lambda_n\phi(t_1) + \alpha\phi(t_1) \\ &= (\lambda_n + \alpha)\phi(t_1)\end{aligned}$$

So $\phi_n(t)$ is also an eigenfunction for $K_Y(t_1, t_2)$ with the eigenvalue $\lambda_n + \alpha$.

9.117
6.94 a)

$$\int_{-T}^{T} R_X(t_1,t_2)\phi(t_2)dt_2 = \lambda\phi(t_1)$$

or

$$\begin{aligned}\lambda\phi(t_1) &= \int_{-T}^{T}\sigma^2 e^{-\alpha|t_1-t_2|}\phi(t_2)dt_2 \\ &= \int_{-T}^{t_1}\sigma^2 e^{-\alpha(t_1-t_2)}\phi(t_2)dt_2 + \int_{t_1}^{T}\sigma^2 e^{\alpha(t_1-t_2)}\phi(t_2)dt_2\end{aligned}$$

b)
$$\begin{aligned}\lambda\frac{d\phi(t_1)}{dt_1} &= \sigma^2\phi(t_1) - \int_{-T}^{t_1}\sigma^2\alpha e^{-\alpha(t_1-t_2)}\phi(t_2)dt_2 \\ &\quad -\sigma^2\phi(t_1) + \int_{t_1}^{T}\sigma^2\alpha e^{\alpha(t_1-t_2)\phi(t_2)}dt_2 \\ &= -\alpha\int_{-T}^{t_1}\sigma^2 e^{-\alpha(t_1-t_2)}\phi(t_2)dt_2 + \alpha\int_{t_2}^{T}\sigma^2 e^{\alpha(t_1-t_2)}\phi(t_2)dt_2 \\ \lambda\frac{d^2\phi(t_1)}{dt_1^2} &= -\alpha\sigma^2\phi(t_1) + \alpha^2\int_{-T}^{t_1}\sigma^2 e^{-\alpha(t_1-t_2)}\phi(t_2)dt_2 \\ &\quad -\alpha\sigma^2\phi(t_1) + \alpha^2\int_{t_1}^{T}\sigma^2 e^{-\alpha(t_1-t_2)}\phi(t_2)dt_2 \\ &= -2\alpha\sigma^2\phi(t_1) + \alpha^2\int_{-T}^{T} R_X(t_1,t_2)\phi(t_2)dt_2 \\ &= (-2\alpha\sigma^2 + \lambda\alpha^2)\phi(t_1) \\ \frac{d^2\phi(t_1)}{dt_1^2} &= \frac{\alpha^2(\lambda - 2\frac{\sigma^2}{\alpha})}{\lambda}\phi(t_1)\end{aligned}$$

c) $\dfrac{d^2\phi(t_1)}{dt_1^2} = \dfrac{\alpha^2(2\frac{\sigma^2}{\alpha} - \lambda)}{\lambda}\phi(t_1) = 0$

$$\phi(t_1) = A\sin bt + B\cos bt$$

where

$$b^2 = \frac{\alpha^2(2\frac{\sigma^2}{\alpha} - \lambda)}{\lambda}$$

In order to satisfy orthogonal condition of $\phi(t)$, $\phi(t)$ should be in the form of $A\sin bt$ or $B\cos bt$.

d) $\phi(t) = A\cos bt$. Substitute the $\phi(t)$ into the integral condition.

$$\begin{aligned}\lambda A\cos bt_1 &= \int_{-T}^{T}\sigma^2 e^{-\alpha|t_1-t_2|}A\cos bt_2 dt_2 \\ &= A\sigma^2\int_{-T}^{t_1} e^{-\alpha(t_1-t_2)}\cos bt_2 dt_2 \\ &\quad +A\sigma^2\int_{t_1}^{T} e^{\alpha(t_1-t_2)}\cos bt_2 dt_2 \\ &= A\sigma^2 e^{-\alpha t_1}\int_{-T}^{t_1} e^{\alpha t_2}\cos bt_2 dt_2 \\ &\quad +A\sigma^2 e^{\alpha t_1}\int_{t_1}^{T} e^{-\alpha t_2}\cos bt_2 dt_2\end{aligned}$$

Cancel A on both sides and let $t_1 \to 0$.

$$\begin{aligned}
\frac{\lambda}{\sigma^2} &= \int_{-T}^{0} e^{\alpha t_2}\cos bt_2 dt_2 + \int_{0}^{T} e^{-\alpha t_2}\cos bt_2 dt_2 \\
&= \frac{e^{\alpha t_2}(\alpha\cos bt_2 + b\sin bt_2)|_{-T}^{0}}{\alpha^2+b^2} + \frac{e^{-\alpha t_2}(-\alpha\cos bt_2 + b\sin bt_2)|_{0}^{T}}{\alpha^2+b^2} \\
\alpha^2+b^2 &= \alpha^2 + 2\sigma^2\alpha/\lambda - \alpha^2 = 2\sigma^2\alpha/\lambda \\
\therefore\ 1 &= \frac{\alpha + e^{-\alpha T}(b\sin bT - \alpha\cos bT)}{\alpha} \\
\tan bT &= \alpha/b
\end{aligned}$$

Substitute $\phi(t) = B\sin bt$ into the integral equation, cancel B on both sides,

$$\lambda\sin bt_1 = \int_{-T}^{t_1}\sigma^2 e^{-\alpha(t_1-t_2)}\sin bt_2 dt_2 + \int_{t_1}^{T}\sigma^2 e^{\alpha(t_1-t_2)}\sin bt_2 dt_2$$

Let $t_1 \to T$

$$\begin{aligned}
\frac{\lambda\sin bT}{\sigma^2} &= \int_{-T}^{T} e^{-\alpha T}e^{\alpha t_2}\sin bt_2 dt_2 \\
e^{\alpha T}\frac{\lambda\sin bT}{\sigma^2} &= \frac{e^{\alpha t_2}(+\alpha\sin bt_2 - b\cos bt_2)|_{-T}^{T}}{\alpha^2+b^2} \\
&= \frac{e^{\alpha T}(\alpha\sin bT - b\cos bT) + e^{-\alpha T}(\alpha\sin bT + b\cos bT)}{2\sigma^2\alpha/\lambda} \\
e^{\alpha T}\tan bT &= \frac{e^{\alpha T}(\alpha\tan bT - b)Ee^{-\alpha T}(\alpha\tan bT + b)}{2\alpha}
\end{aligned}$$

Try

$$\tan bT = -b/\alpha$$

$$RHS = \frac{e^{\alpha T}(-b-b)+0}{2\alpha} = e^{\alpha T}\tan bT = LHS$$

So b is the root of $\tan bT = -b/\sigma$.

e)

$$\begin{aligned}
\int_{-T}^{T}\phi(t)\phi^*(t)dt &= 1 \\
\int_{-T}^{T} A\cos bt A\cos bt dt &= 1 \\
\frac{1}{2}A^2\int_{-T}^{T}(1+\cos 2bt)dt &= 1 \\
\frac{1}{2}A^2\cdot 2T - \frac{A^2}{4b}\sin 2bt\Big|_{-T}^{T} &= 1 \\
A^2\left[T - \frac{1}{2b}\sin 2bT\right] &= 1
\end{aligned}$$

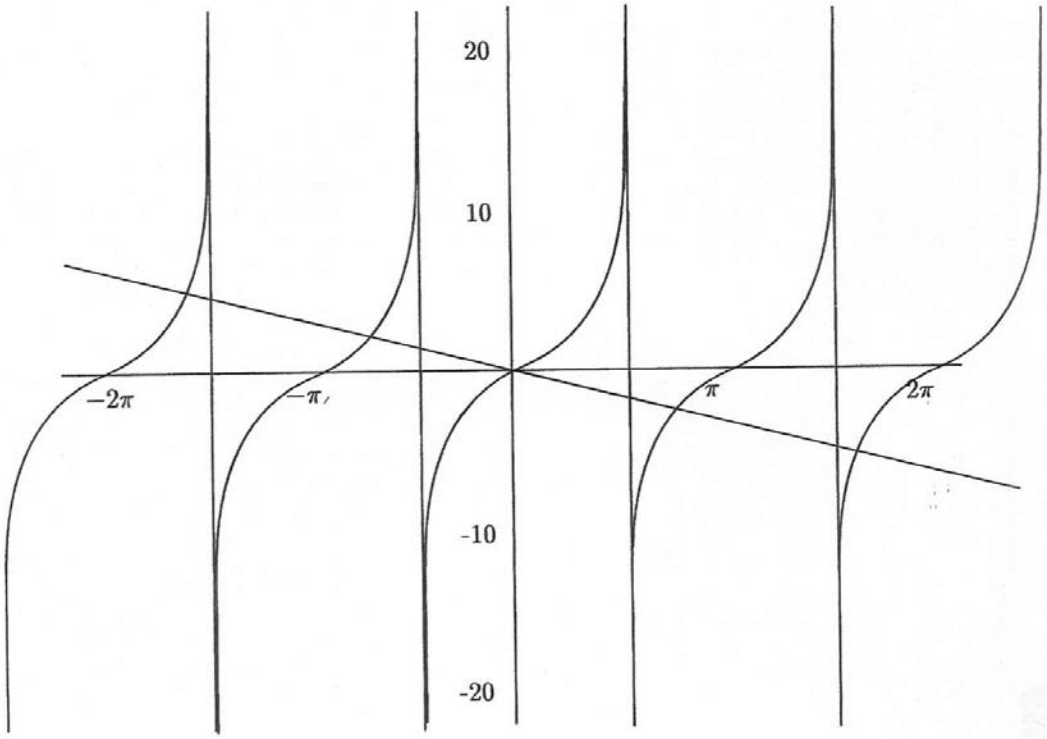
$$A^2 = \frac{1}{T - \frac{1}{2b}\sin 2bT}$$

$$\int_{-T}^{T} B\sin btB\sin b + dt = 1$$
$$\frac{A^2}{2}\int_{-T}^{T}(1-\cos 2bt)dt = 1$$

So

$$B^2 = \frac{1}{T + \frac{1}{2b}\sin 2bT}$$

f)



*9.10 Generating Random Processes

9.118

(a)

```
%P9.118
%part a
clear all;
close all;
s=zeros(200,10,3);
%s dimensions are: (n, realization, p)
p=[0.25 0.5 0.75];
for sample=1:1:10
    for i=1:1:3
        if (rand < p(i))
            s(1,sample,i) = 1;
        end
        for n=2:1:200
            s(n,sample,i)=s(n-1,sample,i);
            if (rand < p(i))
                s(n,sample,i)=s(n-1,sample,i)+1;
            end
        end
    end

    figure(sample);
    plot(1:200,s(:,sample,1),'--',1:200,s(:,sample,2),'-*',
       1:200, s(:,sample,3), '-o');
    legend('p=0.25', 'p=0.5', 'p=0.75');
    xlabel('n')
    ylabel('Sn, random process')
    title('Problem 9.118a');

end
```

(b)

```
%P9.118
%part b
clear all;
close all;
s(1:200,1:50) = 0;
p=0.5;
for sample=1:1:50
    if (rand < p)
        s(1,sample) = 1;
    end
    for n=2:1:200
        s(n,sample)=s(n-1,sample);
        if (rand < p)
            s(n,sample)=s(n-1,sample)+1;
        end
    end
```

```
end
m=mean(s');
v=var(s');
plot(1:200, m(1:200), '--', 1:200, v(1:200), '-o');
legend('mean','variance');
xlabel('n')
ylabel('mean,variance')
      title('Problem 9.118b');
```

(c)

(d)

Octave code for parts c and d:

```
%P9.118
%parts c & d
clear all;
close all;
s(1:200,1:50) = 0;
inc(1:4,1:50) = 0;
p=0.5;
for sample=1:1:50
    if (rand < p)
        s(1,sample) = 1;
    end
    for n=2:1:200
        %{
        %for the distortion case at the end of part d uncomment this part
        %if (n<50)
        %    p=0.9;
        %else
        %    p=0.5;
        %end
        %}
        s(n,sample)=s(n-1,sample);
        if (rand < p)
            s(n,sample)=s(n-1,sample)+1;
        end
    end
    inc(1,sample)=s(50,sample)-s(1,sample);
    inc(2,sample)=s(100,sample)-s(51,sample);
    inc(3,sample)=s(150,sample)-s(101,sample);
    inc(4,sample)=s(200,sample)-s(151,sample);
end

figure(1);
subplot(2,2,1);
hist(inc(1,:),5);
```

```
xlabel('increments [1-50]');
ylabel('number of samples');
subplot(2,2,2);
hist(inc(2,:),5);
xlabel('increments [51-100]');
ylabel('number of samples');
subplot(2,2,3);
hist(inc(3,:),5);
xlabel('increments [101-150]');
ylabel('number of samples');
subplot(2,2,4);
hist(inc(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
%hist(inc',5);

figure(2);
plot(inc(1,:),inc(2,:),'*');
xlabel('inc in [1,50]');
ylabel('inc in [51,100]');
axis([1 50 1 50]);
title('Problem 9.118d');

%for test we can distort inc in one range and see if it affects increments
%in the other range, for example we can modify the parameter p for range
%(1,50) and change it back to the original value for the range (51,100)
```

9.119

(a)

```
clear all;
close all;
s=zeros(3,200,10);
p=[0.25 0.5 0.75];
for sample=1:1:10
    for i=1:1:3
        if (rand < p(i))
            s(i,1,sample) = 1;
        else
            s(i,1,sample) = -1;
        end
        for n=2:1:200
            s(i,n,sample)=s(i,n-1,sample);
            if (rand < p(i))
                s(i,n,sample)=s(i,n-1,sample)+1;
            else
                s(i,n,sample)=s(i,n-1,sample)-1;
            end
        end
    end

    figure(sample);
    plot(1:200, s(1,:,sample), '--', 1:200,s(2,:,sample), '*',
      1:200,s(3,:,sample), 'o');
    legend('p=0.25', 'p=0.5', 'p=0.75');
    xlabel('n')
    ylabel('Sn, random process')
    title('Problem 9.119a');

end
```

(b)

```
clear all;
close all;
s(1:200,1:500) = 0;
p=0.5;
for sample=1:1:50
    if (rand < p)
        s(1,sample) = 1;
    else
        s(1,sample) = -1;
    end
    for n=2:1:200
        s(n,sample)=s(n-1,sample);
        if (rand < p)
            s(n,sample)=s(n-1,sample)+1;
        else
            s(n,sample)=s(n-1,sample)-1;
        end
    end
```

```
end
m=mean(s');
v=var(s');
plot(1:1:200, m, '--', 1:1:200, v, '-o');
legend('mean','variance');
xlabel('n')
ylabel('mean,variance')
title('Problem 9.119b');
```

(c)

(d)

Octave code for parts c and d:

```
clear all;
close all;
s(1:200,1:50) = 0;
inc(1:4,1:50) = 0;
p=0.5;
for sample=1:1:50
    if (rand < p)
        s(1,sample) = 1;
    else
        s(1,sample) = -1;
    end
    for n=2:1:200
        %{
        %for the distortion case at the end of part d uncomment this part
        %if (n<50)
        %    p=0.9;
        %else
        %    p=0.5;
        %end
        %}
        s(n,sample)=s(n-1,sample);
        if (rand < p)
            s(n,sample)=s(n-1,sample)+1;
        else
            s(n,sample)=s(n-1,sample)-1;
        end
    end
    %increment values:
    inc(1,sample)=s(50,sample)-s(1,sample);
    inc(2,sample)=s(100,sample)-s(51,sample);
    inc(3,sample)=s(150,sample)-s(101,sample);
    inc(4,sample)=s(200,sample)-s(151,sample);
end
close all;
figure(1);
subplot(2,2,1);
hist(inc(1,:),5);
xlabel('increments [1-50]');
ylabel('number of samples');
```

```
subplot(2,2,2);
hist(inc(2,:),5);
xlabel('increments [51-100]');
ylabel('number of samples');
subplot(2,2,3);
hist(inc(3,:),5);
xlabel('increments [101-150]');
ylabel('number of samples');
subplot(2,2,4);
hist(inc(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');

%hist(inc',5);
figure(2);
plot(inc(1,:),inc(2,:),'*');
xlabel('inc in [1,50]');
ylabel('inc in [51,100]');
axis([1 50 1 50]);
title('Problem 9.119d');

%for test we can distort inc in one range and see if it affects increments
%in the other range, for example we can modify the parameter p for range
%(1,50) and change it back to the original value for the range (51,100)
```

9.120

(a)

```
clear all;
close all;
s=zeros(2,200,10);
p=[0 0.5];
for sample=1:1:10
    for i=1:1:2
        s(i,1,sample) = 0;
        for n=2:1:200
            s(i,n,sample)=s(i,n-1,sample)+randn+p(i);
        end
    end

    figure(sample);
    plot(1:1:200, s(1,:,sample), '--', 1:1:200, s(2,:,sample), '-*');
    legend('m=0', 'm=0.5');
    xlabel('n')
    ylabel('Sn, random process')
    title('Problem 9.120a');

end
```

(b)

```
clear all;
close all;
s(1:200,1:50) = 0;
p=0.5;
for sample=1:1:50
    s(1,sample) = 0;
    for n=2:1:200
        s(n,sample)=s(n-1,sample)+randn+p;
    end
end
m=mean(s');
v=var(s');
plot(1:200, m, '--', 1:200, v, '-o');
legend('mean','variance');
xlabel('n')
ylabel('mean,variance')
title('Problem 9.120b');
```

(c)

(d)

Octave code for parts c and d:

```
%for the test in part d, we can distort inc in one range and see if it
affects increments
```

```
%in the other range, for example we can modify the parameter p for range
%(1,50) and change it back to the original value for the range (51,100)

clear all;
close all;
s(1:200,1:50) = 0;
inc(1:4,1:50) = 0;
p=0.5;
for sample=1:1:50
    s(1,sample) = 0;
    for n=2:1:200
        %{
        %for the distortion case at the end of part d uncomment this part
        %if (n<50)
        %    p=0.9;
        %else
        %    p=0.5;
        %end
        %}
        s(n,sample)=s(n-1,sample)+randn+p;
    end
    %increment values in each realization:
    inc(1,sample)=s(50,sample)-s(1,sample);
    inc(2,sample)=s(100,sample)-s(51,sample);
    inc(3,sample)=s(150,sample)-s(101,sample);
    inc(4,sample)=s(200,sample)-s(151,sample);
end
figure(1);
subplot(2,2,1);
hist(inc(1,:),5);
xlabel('increments [1-50]');
ylabel('number of samples');
subplot(2,2,2);
hist(inc(2,:),5);
xlabel('increments [51-100]');
ylabel('number of samples');
subplot(2,2,3);
hist(inc(3,:),5);
xlabel('increments [101-150]');
ylabel('number of samples');
subplot(2,2,4);
hist(inc(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
%hist(inc',5);


figure(2);
plot(inc(1,:),inc(2,:),'*');
title('Problem 9.120d');
xlabel('inc in [1,50]');
ylabel('inc in [51,100]');
axis([1 100 1 100]);
```

9.121

(a)

```
clear all;
close all;
s(1:200,1:10) = 0;
for sample=1:1:10
    s(1,sample) = poisson_inv(rand,1);
    for n=2:1:200
        s(n,sample)=s(n-1,sample)+poisson_inv(rand,1);
    end

    figure(sample);
    plot(1:200, s(:,sample));
    xlabel('n')
    ylabel('Sn, random process')
    title('Problem 9.121a');

end
```

(b)

```
clear all;
close all;
s(1:200,1:50) = 0;
for sample=1:1:50
    s(1,sample) = poisson_inv(rand,1);
    for n=2:1:200
        s(n,sample)=s(n-1,sample)+poisson_inv(rand,1);
    end
end
m=mean(s');
v=var(s');
plot(1:200, m, '--', 1:200, v, '-o');
legend('mean','variance');
xlabel('n')
ylabel('mean,variance')
title('Problem 9.121b');
```

(c)

(d)

Octave code for parts c and d:

```
clear all;
close all;
s(1:200,1:50) = 0;
inc(1:4,1:50) = 0;
p=1;
for sample=1:1:50
    s(1,sample) = poisson_inv(rand,p);%poissinv(rand, p);
    for n=2:1:200
```

```
        %{
        %for the distortion case at the end of part d uncomment this part
        %if (n<50)
        %    p=0.9;
        %else
        %    p=0.5;
        %end
        %}
        s(n,sample)=s(n-1,sample)+poisson_inv(rand,p);%poissinv(rand, p);
    end
    inc(1,sample)=s(50,sample)-s(1,sample);
    inc(2,sample)=s(100,sample)-s(51,sample);
    inc(3,sample)=s(150,sample)-s(101,sample);
    inc(4,sample)=s(200,sample)-s(151,sample);
end

%hist(inc',5);
figure(1);
subplot(2,2,1);
hist(inc(1,:),5);
xlabel('increments [1-50]');
ylabel('number of samples');
subplot(2,2,2);
hist(inc(2,:),5);
xlabel('increments [51-100]');
ylabel('number of samples');
subplot(2,2,3);
hist(inc(3,:),5);
xlabel('increments [101-150]');
ylabel('number of samples');
subplot(2,2,4);
hist(inc(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
title('Problem 9.121c');

figure(2);
plot(inc(1,:),inc(2,:), '*');
xlabel('inc in [1,50]');
ylabel('inc in [51,100]');
axis([1 50 1 50]);
title('Problem 9.121d');

%for test we can distort inc in one range and see if it affects increments
%in the other range, for example we can modify the parameter p for range
%(1,50) and change it back to the original value for the range (51,100)
```

9.122

(a)

```
%Cauchy random variable with parameter 1 has CDF of arctan (x)/pi+0.5
clear;
s(1:200,1:10) = 0;
for sample=1:1:10
    s(1,sample) = tan(pi*(rand-0.5));
    for n=2:1:200
        s(n,sample)=s(n-1,sample)+tan(pi*(rand-0.5));
    end

    figure(sample);
    plot(1:1:200, s(:,sample));
    xlabel('n')
    ylabel('Sn, random process')
    title('Problem 9.122a');
end
```

(b)

```
clear all;
close all;
s(1:200,1:50) = 0;
for sample=1:1:50
    s(1,sample) = tan(pi*(rand-0.5));
    for n=2:1:200
        s(n,sample)=s(n-1,sample)+tan(pi*(rand-0.5));
    end
end
m=mean(s');
v=var(s');
plot(1:1:200, m, '--', 1:1:200, v, 'o');
legend('mean','variance');
xlabel('n')
ylabel('mean,variance')
title('Problem 9.122b');
```

(c)

(d)

Octave code for parts c and d:

```
%for test in part d, we can distort inc in one range and see if it affects
increments
%in the other range, for example we can modify the parameter p for range
%(1,50) and change it back to the original value for the range (51,100)

clear all;
close all;
```

```
s(1:200,1:50) = 0;
inc(1:4,1:50) = 0;
for sample=1:1:50
    s(1,sample) = tan(pi*(rand-0.5));
    for n=2:1:200
        s(n,sample)=s(n-1,sample)+tan(pi*(rand-0.5));
        %{
        %for the distortion case at the end of part d uncomment this part
        %if (n<50)
        %    s(n,sample)=s(n-1,sample)+tan(pi*(rand/2-0.5));
        %end
        %}
    end
    inc(1,sample)=s(50,sample)-s(1,sample);
    inc(2,sample)=s(100,sample)-s(51,sample);
    inc(3,sample)=s(150,sample)-s(101,sample);
    inc(4,sample)=s(200,sample)-s(151,sample);
end

%hist(inc',5);
figure(1);
subplot(2,2,1);
hist(inc(1,:),5);
xlabel('increments [1-50]');
ylabel('number of samples');
subplot(2,2,2);
hist(inc(2,:),5);
xlabel('increments [51-100]');
ylabel('number of samples');
subplot(2,2,3);
hist(inc(3,:),5);
xlabel('increments [101-150]');
ylabel('number of samples');
subplot(2,2,4);
hist(inc(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
title('Problem 9.122c');

figure(2);
plot(inc(1,:),inc(2,:), '*');
xlabel('inc in [1,50]');
ylabel('inc in [51,100]');
axis([1 100 1 100]);
title('Problem 9.122d');
```

9.123

(a)

```
clear all;
close all;
y=zeros(5,200,3,2);
%dimensions in y are:  (realization, n, alpha, p)
alpha=[0.25 0.5 0.9];
step=0;
p=[0.5 0.25];
for sample=1:1:5
    for i=1:1:3
        for j=1:1:2
            rn=rand;
            step = -1*(rn <= p(j))+1*(rn > p(j));
            y(sample,1,i,j)=step;
            for n=2:1:200
                rn=rand;
                step = -1*(rn <= p(j))+1*(rn > p(j));
                y(sample,n,i,j)=alpha(i)*y(sample,n-1,i,j)+step;
            end
        end

        figure(sample*4+i);
        plot(1:200, y(sample,1:200,i,1), '--', 1:200, y(sample,1:200,i,2));
        legend('p=0.5', 'p=0.25');
        xlabel('n')
        ylabel('Yn, random process')
        str=sprintf('Problem 9.123a, alpha=%1.1f',alpha(i));
        title(str);

    end
end
m=mean(y);
v=var(y);
%plotting mean and variance
for i=1:1:3
    figure(200+i);
    subplot(2,1,1);
    plot(1:1:200, m(1,1:200,i,1), '--', 1:1:200, m(1,1:200,i,2));
    legend('p=0.5', 'p=0.25');
    xlabel('n')
    ylabel('mean of Yn')
    str=sprintf('Problem 9.123a, alpha=%1.1f',alpha(i));
    title(str);
    subplot(2,1,2);
    plot(1:200, v(1,1:200,i,1), '--', 1:200, v(1,1:200,i,2));
    legend('p=0.5', 'p=0.25');
    xlabel('n')
    ylabel('variance of Yn')
    title(str);
end
```

```
%histogram
for sample=1:1:5
    for i=1:1:3
        figure(300+sample*4+i);
        for j=1:1:2
            subplot(2,1,j);
            hist(y(sample,:,i,j));
            xlabel('Yn, output')
            ylabel('Histogram count')
            str=sprintf('Problem 9.121a, histogram for alpha=%1.1f,
               p=%1.1f,sample#%d',alpha(i),p(j),sample);
            title(str);
        end
    end
end
```

(b)

```
clear all;
close all;
y=zeros(50,200,2);
%dimensions are: (realization, n, p)
alpha=0.5;
step=0;
p=[0.5 0.25];
for sample=1:1:50
    for j=1:1:2
        rn = rand;
        step = -1*(rn <= p(j))+1*(rn > p(j));
        y(sample,1,j)=step;
        for n=2:1:200
            rn = rand;
            step = -1*(rn <= p(j))+1*(rn > p(j));
            y(sample,n,j)=alpha*y(sample,n-1,j)+step;
        end
    end
end

m=mean(y);
v=var(y);
%plotting mean and variance:
figure(100);
subplot(2,1,1);
plot(1:1:200, m(1,1:200,1), '--', 1:1:200, m(1,1:200,2));
legend('p=0.5', 'p=0.25');
xlabel('n')
ylabel('mean of Yn')
str=sprintf('Problem 9.123b, alpha=%f',alpha);
title(str);
subplot(2,1,2);
plot(1:1:200, v(1,1:200,1), '--', 1:1:200, v(1,1:200,2));
legend('p=0.5', 'p=0.25');
xlabel('n')
ylabel('variance of Yn')
title(str);
figure;
```

```
%histogram
figure(200);
for j=1:1:2
    subplot(2,3,(j-1)*3+1);
    hist(y(:,5,j));
    xlabel('outcome')
    str=sprintf('p=%f',p(j));
    ylabel(str)
    str=sprintf('P.9.123b, n=5');
    title(str);
    subplot(2,3,(j-1)*3+2);
    hist(y(:,50,j));
    str=sprintf('P.9.123b, n=50');
    title(str);
    subplot(2,3,(j-1)*3+3);
    hist(y(:,200,j));
    str=sprintf('P.9.123b, n=200');
    title(str);
end
```

(c)

```
clear all;
close all;
y=zeros(50,200,2);
%y dimensions: (realization, n, p)
inc1(1:4,1:50)=0;
inc2(1:4,1:50)=0;
alpha=0.5;
step=0;
p=[0.5 0.25];
for sample=1:1:50
    for j=1:1:2
        rn=rand;
        step = -1*(rn <= p(j))+1*(rn > p(j));
        y(sample,1,j)=step;
        for n=2:1:200
            rn = rand;
            step = -1*(rn <= p(j))+1*(rn > p(j));
            y(sample,n,j)=alpha*y(sample,n-1,j)+step;
        end
    end
    inc1(1,sample)=y(sample,50,1)-y(sample,1,1);
    inc1(2,sample)=y(sample,100,1)-y(sample,51,1);
    inc1(3,sample)=y(sample,150,1)-y(sample,101,1);
    inc1(4,sample)=y(sample,200,1)-y(sample,151,1);
    inc2(1,sample)=y(sample,50,2)-y(sample,1,2);
    inc2(2,sample)=y(sample,100,2)-y(sample,51,2);
    inc2(3,sample)=y(sample,150,2)-y(sample,101,2);
    inc2(4,sample)=y(sample,200,2)-y(sample,151,2);
end

%hist(inc1',5);
figure(1);
subplot(2,2,1);
```

```
hist(inc1(1,:),5);
xlabel('increments [1-50]');
ylabel('number of samples');
title('Problem 9.123c, p=0.5');
subplot(2,2,2);
hist(inc1(2,:),5);
xlabel('increments [51-100]');
ylabel('number of samples');
title('Problem 9-121-c, p=0.5');
subplot(2,2,3);
hist(inc1(3,:),5);
xlabel('increments [101-150]');
ylabel('number of samples');
title('Problem 9.123c, p=0.5');
subplot(2,2,4);
hist(inc1(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
title('Problem 9.123c, p=0.5');
replot;

figure(2);
%hist(inc2',5);
subplot(2,2,1);
hist(inc2(1,:),5);
title('Problem 9.123c, p=0.25');
xlabel('increments [1-50]');
ylabel('number of samples');
subplot(2,2,2);
hist(inc2(2,:),5);
title('Problem 9.123c, p=0.25');
xlabel('increments [51-100]');
ylabel('number of samples');
subplot(2,2,3);
hist(inc2(3,:),5);
title('Problem 9.123c, p=0.25');
xlabel('increments [101-150]');
ylabel('number of samples');
subplot(2,2,4);
hist(inc2(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
title('Problem 9.123c, p=0.25');
replot
```

9.124

(a)

```
clear all;
close all;
y=zeros(5,200,2);
%y dimensions: (realization, n, gaussian mean)
step=0;
p=[0 0.5];
for sample=1:1:5
    for j=1:1:2
        step = randn+p(j);
        y(sample,1,j)=step;
        for n=2:1:200
            step = randn+p(j);
            y(sample,n,j)=y(sample,n-1,j)+step;
        end
    end

    figure(sample);
    plot(1:200, y(sample,1:200,1), 1:200, y(sample,1:200,2));
    legend('m=0', 'm=0.5');
    xlabel('n')
    ylabel('Yn, random process')
    title('Problem 9.124a');
end
m=mean(y);
v=var(y);
figure(100);
subplot(2,1,1);
plot(1:200, m(1,1:200,1), '--', 1:200, m(1,1:200,2));
legend('m=0', 'm=0.5');
xlabel('n')
ylabel('mean of Yn')
title('Problem 9.124a');
subplot(2,1,2);
plot(1:1:200, v(1,1:200,1), '--', 1:1:200, v(1,1:200,2));
legend('m=0.5', 'm=0.25');
xlabel('n')
ylabel('variance of Yn')
title('Problem 9.124a');

%histogram
for sample=1:1:5
    figure(200+sample);
    for j=1:1:2
        subplot(2,1,j);
        hist(y(sample,1:200,j));
        xlabel('n, number of trials')
        ylabel('Histogram count')
        str=sprintf('Problem 9.124a, histogram for
m=%1.1f,sample#%d',p(j),sample);
        title(str);
    end
end
```

(b)

```
clear all;
close all;
y=zeros(50,200,2);
step=0;
p=[0 0.5];
for sample=1:1:50
    for j=1:1:2
        step = randn+p(j);
        y(sample,1,j)=step;
        for n=2:1:200
            step = randn+p(j);
            y(sample,n,j)=y(sample,n-1,j)+step;
        end

    end
end
m=mean(y);
v=var(y);
figure(1);
subplot(2,1,1);
plot(1:200, m(1,1:200,1), '--', 1:200, m(1,1:200,2));
legend('m=0', 'm=0.5');
xlabel('n')
ylabel('mean of Yn')
title('Problem 9-122-b');
subplot(2,1,2);
plot(1:1:200, v(1,1:200,1), '--', 1:1:200, v(1,1:200,2));
legend('m=0', 'm=0.5');
xlabel('n')
ylabel('variance of Yn')
title('Problem 9.124b');


%histogram

figure(100);
for j=1:1:2
    subplot(2,3,(j-1)*3+1);
    hist(y(:,5,j));
    xlabel('outcome')
    str=sprintf('m=%1.1f',p(j));
    ylabel(str)
    str=sprintf('P9.124b, n=5');
    title(str);
    subplot(2,3,(j-1)*3+2);
    hist(y(:,50,j));
    str=sprintf('P9.124b, n=50');
    title(str);
    subplot(2,3,(j-1)*3+3);
    hist(y(:,200,j));
    str=sprintf('P9.124b, n=200');
    title(str);
end
```

(c)

```
clear;
close all;
y=zeros(50,200,2);
step=0;
p=[0 0.5];
for sample=1:1:50
    for j=1:1:2
        step = randn+p(j);
        y(sample,1,j)=step;
        for n=2:1:200
            step = randn+p(j);
            y(sample,n,j)=y(sample,n-1,j)+step;
        end
    end
    inc1(1,sample)=y(sample,50,1)-y(sample,1,1);
    inc1(2,sample)=y(sample,100,1)-y(sample,51,1);
    inc1(3,sample)=y(sample,150,1)-y(sample,101,1);
    inc1(4,sample)=y(sample,200,1)-y(sample,151,1);
    inc2(1,sample)=y(sample,50,2)-y(sample,1,2);
    inc2(2,sample)=y(sample,100,2)-y(sample,51,2);
    inc2(3,sample)=y(sample,150,2)-y(sample,101,2);
    inc2(4,sample)=y(sample,200,2)-y(sample,151,2);
end

figure(1);
subplot(2,2,1);
hist(inc1(1,:),5);
xlabel('increments [1-50]');
ylabel('number of samples');
title('Problem 9.124c, m=0');
subplot(2,2,2);
hist(inc1(2,:),5);
xlabel('increments [51-100]');
ylabel('number of samples');
title('Problem 9.124c, m=0');
subplot(2,2,3);
hist(inc1(3,:),5);
xlabel('increments [101-150]');
ylabel('number of samples');
title('Problem 9.124c, m=0');
subplot(2,2,4);
hist(inc1(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
title('Problem 9.124c, m=0');
replot;

figure(2);
%hist(inc2',5);
subplot(2,2,1);
hist(inc2(1,:),5);
title('Problem 9.124c, m=0.5');
xlabel('increments [1-50]');
ylabel('number of samples');
```

```
subplot(2,2,2);
hist(inc2(2,:),5);
title('Problem 9.124c, m=0.5');
xlabel('increments [51-100]');
ylabel('number of samples');
subplot(2,2,3);
hist(inc2(3,:),5);
title('Problem 9.124c, m=0.5');
xlabel('increments [101-150]');
ylabel('number of samples');
subplot(2,2,4);
hist(inc2(4,:),5);
xlabel('increments [151-200]');
ylabel('number of samples');
title('Problem 9.124c, m=0.5');
replot
```

9.125

Octave code for parts a, b and c:

```
clear all;
close all;
s(1:200,1:50) = 0;
p=0.5;
for sample=1:1:50
    if (rand < p)
        s(1,sample) = 1;
    end
    for n=2:1:200
        s(n,sample)=s(n-1,sample);
        if (rand < p)
            s(n,sample)=s(n-1,sample)+1;
        end
    end
end
m=mean(s');
v=var(s');

%For auto correlation correlation:
%acor=s*s'/number of samples
%for autocovariance: acov=acor-m*m';

acor=s*s'/50;
acov=acor-m'*m;

%for WSS, mean and variance should not depend on (n)
%so for P9.125c since variance depends on (n) it is not WSS.

plot(1:200, m, '--', 1:200, v, 'o');
legend('mean','variance');
xlabel('n')
ylabel('mean,variance')
title('Problem 9.123c');
```

(d)

Octave code for parts a, b and d:

```
clear all;
close all;
numberofsamples=1000;
y1=zeros(numberofsamples,200);
alpha=0.5;
step=0;
p=0.5;
for sample=1:1:numberofsamples
    rn=rand;
    step = -1*(rn <= p)+1*(rn > p);
    y1(sample,1)=step;
    for n=2:1:200
        rn = rand;
```

```
        step = -1*(rn <= p)+1*(rn > p);
        y1(sample,n)=alpha*y1(sample,n-1)+step;
    end
end
m1=mean(y1);
v1=var(y1);


%For auto correlation correlation:
%acor=y*y'/number of samples
%for autocovariance: acov=acor-m'*m;


acor1=y1'*y1/numberofsamples;
acov1=acor1-m1'*m1;

%for WSS, mean and variance should not depend on (n), and acor should
%depend only on (n1-n2)
for i=1:1:200
    v12(i)=acov1(i,i);
end

figure (1);
plot(1:1:200, m1, '--', 1:1:200, v1, '-', 1:200, v12, '*');
legend('mean', 'variance', 'var from acov');
xlabel('n')
ylabel('mean of Yn, p=0.5')
title('Problem 9-123-d');

y2=zeros(numberofsamples,200);
step=0;
p=0.25;
for sample=1:1:numberofsamples
    rn = rand;
    step = -1*(rn <= p)+1*(rn > p);
    y2(sample,1)=step;
    for n=2:1:200
        rn = rand;
        step = -1*(rn <= p)+1*(rn > p);
        y2(sample,n)=alpha*y2(sample,n-1)+step;
    end
end
m2=mean(y2);
v2=var(y2);

%For auto correlation correlation:
%acor=y*y'/number of samples
%for autocovariance: acov=acor-m'*m;

acor2=y2'*y2/numberofsamples;
acov2=acor2-m2'*m2;

%for WSS, mean and variance should not depend on (n), and acor should
%depend only on (n1-n2)
for i=1:1:200
    v22(i)=acov2(i,i);
end
```

```
figure (2);
plot(1:200, m2, '--', 1:200, v2, '-', 1:200, v22, '*');
legend('mean', 'var', 'var from acov');
xlabel('n')
ylabel('mean of Yn, p=0.25')
title('Problem 9.125d');

figure (3);
mesh(acov1);
figure (4);
mesh(acov2);

%you can see that v22 & v12 are the variances computed from autocovariance
%functions, now if you draw the autocovariance, you can see that it is
%almost zero for n1-n2 <> 0
%also it can be seen that mean and variance are independent of n
```

9.126

```
%lambda*100=n*p, if lambda=1 then n*p should be 100
clear all;
close all;
%as n grows N would be a better approximation of a Poisson process.
%probably n=10*100 would be a good pick
%you can draw N for n large than 10*t and you can see that the result would
%not change significantly

n=1000;
p=100/n;
N(1:n)=0;
N(1)=0;
for i=2:1:n
    if (rand < p)
        N(i)=N(i-1)+1;
    else
        N(i)=N(i-1);
    end
end

plot(N);
```

9.127

Octave code for parts a and b:

```
clear all;
close all;
%lambda is 1/4 per second
%lamda*t=n*p=60*1/4=15
n=1200;
p=15/n;
rep=100;
cnt = 0;
dur(1:rep)=0;
for r=1:1:rep
    N(1:1:n)=0;
    N(1)=0;
    arr = 0;
    for i=2:1:n
        if (rand < p)
            arr = arr + 1;
            if (arr == 1)
                previ=i;
            end
            if (arr == 2)
                dur(r) = i-previ;
            end
            N(i)=N(i-1)+1;
        else
            N(i)=N(i-1);
        end
    end
    if ((N(10*n/60) == 3) && ((N(60*n/60)-N(45*n/60)) == 2))
        cnt = cnt + 1;
    end
end
disp('relative frequency');
cnt/rep
disp('theoretical prob');
(10/4)^3*(15/4)^2*exp(-10/4-15/4)/12
figure(1);
hist(dur,1:n/60:40*n/60);
axis([0 40*n/60]);
figure(2);
z=hist(dur,1:n/60:40*n/60);
%pdf is exponential with rate lambda
t=1:1:40;
f=0.25*exp(-0.25*t);
plot(t,f, '-', t,z/rep, '--');
legend('theoretical', 'simulation');
xlabel('seconds');
title('pdf');
figure(3);
cf=1-exp(-0.25*t);
plot(t,cf, '-', t,cumsum(z)/rep, '--');
legend('theoretical', 'simulation');
xlabel('seconds');
title('cdf');
```

9.128

(a)

```
%lambda*10=n*p, if lambda=1 then n*p should be 10
clear all;
close all;
n=100;
p=10/n;
maxrep = 100;
N(1:maxrep,1:n)=0;
N1(1:maxrep,1:n)=0;
N2(1:maxrep,1:n)=0;
for r=1:1:maxrep
    N(r,1)=0;
    N1(r,1)=0;
    N2(r,1)=0;
    for i=2:1:n
        if (rand < p)
            N(r,i)=N(r,i-1)+1;
            if (rand < 0.25)
                N1(r,i)=N1(r,i-1)+1;
                N2(r,i)=N2(r,i-1);
            else
                N1(r,i)=N1(r,i-1);
                N2(r,i)=N2(r,i-1)+1;
            end
        else
            N(r,i)=N(r,i-1);
            N1(r,i)=N1(r,i-1);
            N2(r,i)=N2(r,i-1);
        end
    end
end

y=hist(N(:,n),1:21);
y1=hist(N1(:,n),1:21);
y2=hist(N2(:,n),1:21);

%we know that theoretical pmf for N1 and N2 are Poisson random variables
%pmf with rate lambda*p*t, and lambda*(1-p)*t
expected=poisson_pdf(1:21,10);
expected1=poisson_pdf(1:21,10*0.25);
expected2=poisson_pdf(1:21,10*0.75);
figure(1);
plot(1:21,y/maxrep, '-*', 1:21,expected, '-o');
legend('histogram','Poisson');
title('N(t)');
figure(2);
plot(1:21,y1/maxrep, '-*', 1:21,expected1, '-o');
legend('histogram','Poisson');
title('N1(t)');
figure(3);
plot(1:21,y2/maxrep, '-*', 1:21,expected2, '-o');
legend('histogram','Poisson');
title('N2(t)');
```

```
%chi-square goodness-of-fit test: sum(y-expected)^2/expected

p=sum((y/maxrep-expected).^2./expected)
p1=sum((y1/maxrep-expected1).^2./expected1)
p2=sum((y2/maxrep-expected2).^2./expected2)



%as n grows N would be a better approximation of a Poisson process.
```

(b)

```
%lambda*10=n*p, if lambda=1 then n*p should be 10
clear all;
close all;
n=100;
p=10/n;
maxrep = 100;
N(1:maxrep,1:1:n)=0;
N1(1:maxrep,1:1:n)=0;
N2(1:maxrep,1:1:n)=0;
for r=1:1:maxrep
    N(r,1)=0;
    N1(r,1)=0;
    N2(r,1)=0;
    for i=2:1:n
        if (rand < p)
            N(r,i)=N(r,i-1)+1;
            if (rand < 0.25)
                N1(r,i)=N1(r,i-1)+1;
                N2(r,i)=N2(r,i-1);
            else
                N1(r,i)=N1(r,i-1);
                N2(r,i)=N2(r,i-1)+1;
            end
        else
            N(r,i)=N(r,i-1);
            N1(r,i)=N1(r,i-1);
            N2(r,i)=N2(r,i-1);
        end
    end
end

y=hist(N(:,n),0:21);
y1=hist(N1(:,n),0:21);
y2=hist(N2(:,n),0:21);

%we know that theoretical pmf for N1 and N2 are Poisson random variables
%pmf with rate lambda*p*t, and lambda*(1-p)*t
expected=poisson_pdf(0:21,10);
expected1=poisson_pdf(0:21,10*0.25);
expected2=poisson_pdf(0:21,10*0.75);
figure(1);
plot(0:21,y/maxrep, '-*', 0:21,expected, '-o');
legend('histogram','poisson');
title('N(t)');
```

```
figure(2);
plot(0:21,y1/maxrep, '-*', 0:21,expected1, '-o');
legend('histogram','poisson');
title('N1(t)');
figure(3);
plot(0:21,y2/maxrep, '-*', 0:21,expected2, '-o');
legend('histogram','poisson');
title('N2(t)');

%chi-square goodness-of-fit test: sum((y-expected)^2/expected)

p=sum((y/maxrep-expected).^2./expected)
p1=sum((y1/maxrep-expected1).^2./expected1)
p2=sum((y2/maxrep-expected2).^2./expected2)

%y/maxrep,y1/maxrep,y2/maxrep are pmf of N,N1, and N2

expec(1:20,1:20)=0;
Freq(1:20,1:20)=0;
for r=1:1:maxrep
    i = N1(r,n)+1;
    j = N2(r,n)+1;
    if (i>20)
        i = 20;
    end
    if (j>15)
        j = 20;
    end
    Freq(i,j)=Freq(i,j)+1;
end
chi = 0;
for i=1:1:19
    for j=1:1:19
       expec(i,j)=expected1(i)*expected2(j);
       chi=chi+(Freq(i,j)/maxrep-expec(i,j))^2/(expec(i,j));
    end
end
chi
figure(10);
subplot(2,1,1);
mesh(expec);
ylabel('theoretical prob');
subplot(2,1,2);
mesh(Freq/maxrep);
ylabel('simulation');
```

9.129

(a)

```
%lambda*t=n*p, if lambda=1 then n*p should be t
clear all;
close all;
n=2000;
p=200/n;
maxrep = 1;
N(1:maxrep,1:1:n)=0;
A(1:maxrep,1:1:n)=0;
D(1:maxrep,1:1:n)=0;
L(1:maxrep,1:1:n)=0; % # of customers leaving at time t in (1,n)
for r=1:1:maxrep
    %assuming we will have maximum of 3000 customer and their service time
distribution is exponential.
    T(1:1:3000) = floor(exponential_inv(rand(1,3000), 5*n/200));
    N(r,1)=0;
    A(r,1)=0;
    D(r,1)=0;
    cust=1;
    for i=2:1:n
        if (rand < p)
            A(r,i)=A(r,i-1)+1;
            N(r,i)=N(r,i-1)+1;
            if (i+T(cust) < n)
                L(r,i+T(cust))=L(r,i+T(cust))+1;
            end
            cust = cust+1;
        else
            A(r,i)=A(r,i-1);
            N(r,i)=N(r,i-1);
        end
        D(r,i)=D(r,i-1)+L(r,i);
        N(r,i)=N(r,i)-L(r,i);
    end
end

plot(1:1:n,A(1,:),1:1:n,N(1,:),1:1:n,D(1,:));
legend('Arrival','Number in system','Departure');
```

(b)

(c)

First part of code is the same as part a.

```
%lambda*t=n*p, if lambda=1 then n*p should be 200
clear all;
close all;
n=2000;
p=200/n;
t=n/200; % one second is t steps
```

```
maxrep = 100;
N(1:maxrep,1:1:n)=0;
A(1:maxrep,1:1:n)=0;
D(1:maxrep,1:1:n)=0;
L(1:maxrep,1:1:n)=0; % # of customers leaving at time t in (1,n)
for r=1:1:maxrep
    %assuming we will have maximum of 3000 customer and their service time
distribution is exponential.
    T(1:1:3000) = floor(exponential_inv(rand(1,3000), 5*t));
    N(r,1)=0;
    A(r,1)=0;
    D(r,1)=0;
    cust=1;
    for i=2:1:n
        if (rand < p)
            A(r,i)=A(r,i-1)+1;
            N(r,i)=N(r,i-1)+1;
            if (i+T(cust) < n)
                L(r,i+T(cust))=L(r,i+T(cust))+1;
            end
            cust = cust+1;
        else
            A(r,i)=A(r,i-1);
            N(r,i)=N(r,i-1);
        end
        D(r,i)=D(r,i-1)+L(r,i);
        N(r,i)=N(r,i)-L(r,i);
    end
end

figure(1);
plot(1:1:n,A(1,:),1:1:n,N(1,:),1:1:n,D(1,:));
legend('Arrival','Number in system','Departure');

d50=hist(D(:,50*t),1:max(D(:,50*t)))/maxrep;
d100=hist(D(:,100*t),1:max(D(:,100*t)))/maxrep;
d150=hist(D(:,150*t),1:max(D(:,150*t)))/maxrep;
d200=hist(D(:,200*t),1:max(D(:,200*t)))/maxrep;

%we know that theoretical pmf for D(t) is poisson random variables
%pmf with rate lambda*(t-5)

expected50=poisson_pdf(1:max(D(:,50*t)), 45);
expected100=poisson_pdf(1:max(D(:,100*t)), 95);
expected150=poisson_pdf(1:max(D(:,150*t)), 145);
expected200=poisson_pdf(1:max(D(:,200*t)), 195);

figure(2);
plot(1:max(D(:,50*t)),d50, '-*', 1:max(D(:,50*t)),expected50, '-o');
legend('histogram','poisson');
title('D(50)');

figure(3);
plot(1:max(D(:,100*t)),d100, '-*', 1:max(D(:,100*t)),expected100, '-o');
legend('histogram','poisson');
title('D(100)');
figure(4);
```

```
plot(1:max(D(:,150*t)),d150, '-*', 1:max(D(:,150*t)),expected150, '-o');
legend('histogram','poisson');
title('D(150)');
figure(5);
plot(1:max(D(:,200*t)),d200, '-*', 1:max(D(:,200*t)),expected200, '-o');
legend('histogram','poisson');
title('D(200)');


%chi-square goodness-of-fit test: sum(y-expected)^2/expected

p50=sum((d50-expected50).^2./expected50)
p100=sum((d100-expected100).^2./expected100)
p150=sum((d150-expected150).^2./expected150)
p200=sum((d200-expected200).^2./expected200)
```

(d)

```
%lambda*t=n*p, if lambda=1 then n*p should be 200 seconds
clear all;
close all;
n=2000;
p=200/n;
t=n/200; % one second is t steps
maxrep = 100;
N(1:maxrep,1:1:n)=0;
A(1:maxrep,1:1:n)=0;
D(1:maxrep,1:1:n)=0;
L(1:maxrep,1:1:n)=0; % # of customers leaving at time t in (1,n)
for r=1:1:maxrep
    N(r,1)=0;
    A(r,1)=0;
    D(r,1)=0;
    cust=1;
    for i=2:1:n
        if (rand < p)
            A(r,i)=A(r,i-1)+1;
            N(r,i)=N(r,i-1)+1;
            if (i+5*t < n)
                L(r,i+5*t)=L(r,i+5*t)+1;
            end
            cust = cust+1;
        else
            A(r,i)=A(r,i-1);
            N(r,i)=N(r,i-1);
        end
        D(r,i)=D(r,i-1)+L(r,i);
        N(r,i)=N(r,i)-L(r,i);
    end
end

figure(1);
plot(1:1:n,A(1,:),1:1:n,N(1,:),1:1:n,D(1,:));
legend('Arrival','Number in system','Departure');

d50=hist(D(:,50*t),1:max(D(:,50*t)))/maxrep;
```

```
d100=hist(D(:,100*t),1:max(D(:,100*t)))/maxrep;
d150=hist(D(:,150*t),1:max(D(:,150*t)))/maxrep;
d200=hist(D(:,200*t),1:max(D(:,200*t)))/maxrep;

%we know that theoretical pmf for D(t) is poisson random variables
%pmf with rate lambda*(t-5)

expected50=poisson_pdf(1:max(D(:,50*t)), 45);
expected100=poisson_pdf(1:max(D(:,100*t)), 95);
expected150=poisson_pdf(1:max(D(:,150*t)), 145);
expected200=poisson_pdf(1:max(D(:,200*t)), 195);

figure(2);
plot(1:max(D(:,50*t)),d50, '-*', 1:max(D(:,50*t)),expected50, '-o');
legend('histogram','poisson');
title('D(50)');
figure(3);
plot(1:max(D(:,100*t)),d100, '-*', 1:max(D(:,100*t)),expected100, '-o');
legend('histogram','poisson');
title('D(100)');
figure(4);
plot(1:max(D(:,150*t)),d150, '-*', 1:max(D(:,150*t)),expected150, '-o');
legend('histogram','poisson');
title('D(150)');
figure(5);
plot(1:max(D(:,200*t)),d200, '-*', 1:max(D(:,200*t)),expected200, '-o');
legend('histogram','poisson');
title('D(200)');

%chi-square goodness-of-fit test: sum(y-expected)^2/expected

p50=sum((d50-expected50).^2./expected50)
p100=sum((d100-expected100).^2./expected100)
p150=sum((d150-expected150).^2./expected150)
p200=sum((d200-expected200).^2./expected200)
```

9.130

(a)

```
clear all;
close all;
s=100; %each second divided to 100 steps
%alpha=1, so h=sqrt(alpha/s)=0.1
alpha=1;
h=sqrt(alpha/s);
n=s*3.5;
maxrep=1000;
x(1:maxrep,1:n)=0;
inc(1:maxrep,3)=0;
for r=1:1:maxrep
    x(r,1)=0;
    for i=2:1:n
        if (rand < 0.5)
            x(r,i)=x(r,i-1)+h;
        else
            x(r,i)=x(r,i-1)-h;
        end
    end
    inc(r,1)=x(r,0.5*s)-x(r,1);
    inc(r,2)=x(r,1.5*s)-x(r,0.5*s+1);
    inc(r,3)=x(r,3.5*s)-x(r,1.5*s+1);
end

[y b]=hist(inc(:,1),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(0.5));
sno=sum(no);
figure(1);
plot(-6:0.4:6,no/sno,b,y/maxrep)
legend('normal pdf','simulation');
title('inc (0,0.5)');

[y b]=hist(inc(:,2),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(1.5-0.5));
sno=sum(no);
figure(2);
plot(-6:0.4:6,no/sno,b,y/maxrep)
legend('normal pdf','simulation');
title('inc (0.5,1.5)');

[y b]=hist(inc(:,3),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-8:0.4:8,0,sqrt(3.5-1.5));
sno=sum(no);
figure(3);
plot(-8:0.4:8,no/sno,b,y/maxrep)
legend('normal pdf','simulation');
title('inc (3.5,1.5)');
```

(b)

```
clear all;
close all;
s=100; %each second divided to 100 steps
%alpha=1, so h=sqrt(alpha/s)=0.1
alpha=1;
h=sqrt(alpha/s);
n=s*3.5;
maxrep=100;
x(1:maxrep,1:n)=0;
inc(1:maxrep,3)=0;
for r=1:1:maxrep
    x(r,1)=0;
    for i=2:1:n
        if (rand < 0.5)
            x(r,i)=x(r,i-1)+h;
        else
            x(r,i)=x(r,i-1)-h;
        end
    end
    inc(r,1)=x(r,0.5*s)-x(r,1);
    inc(r,2)=x(r,1.5*s)-x(r,0.5*s+1);
    inc(r,3)=x(r,3.5*s)-x(r,1.5*s+1);
end

[y b]=hist(inc(:,1),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(0.5));
sno=sum(no);
figure(1);
plot(-6:0.4:6,no/sno, '--',b,y/maxrep,'-o')
legend('normal','histogram');
title('inc(0,0.5)');
pdf1=no/sno;

[y b]=hist(inc(:,2),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(1.5-0.5));
sno=sum(no);
figure(2);
plot(-6:0.4:6,no/sno, '--',b,y/maxrep,'-o')
legend('normal','histogram');
title('inc(0.5,1.5)');
pdf2=no/sno;

[y b]=hist(inc(:,3),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(3.5-1.5));
sno=sum(no);
pdf3=no/sno;
figure(3);
plot(-6:0.4:6,no/sno, '--',b,y/maxrep,'-o')
legend('normal','histogram');
title('inc(1.5,3.5)');
```

```
value=-6:0.4:6;
vallen=length(value);
%number of values is 12*2.5=30

chi=0;
p0(1:1:vallen,1:1:vallen)=0;
freq(1:1:vallen,1:1:vallen)=0;
for i=2:1:vallen-1
    for j=2:1:vallen-1
        p0(i,j)=pdf1(i)*pdf2(j);
        for r=1:1:maxrep
            if (inc(r,1) <= ((value(i)+value(i+1))/2) && inc(r,1) >
((value(i-1)+value(i))/2))
                if (inc(r,2) <= ((value(j)+value(j+1))/2) && inc(r,2) >
((value(j-1)+value(j))/2))
                    freq(i,j) = freq(i,j)+1;
                end
            end
        end
        freq(i,j)=freq(i,j)/maxrep;
        chi=chi+((freq(i,j)-p0(i,j))^2/p0(i,j));
    end
end
chi %(chi for 10000 sim is 0.02), (chi for 1000 sim is 0.35)
figure(4);
subplot(2,1,1);
mesh(value,value,p0);
axis([-5 5 -5 5]);
zlabel('theoretical prob');
subplot(2,1,2);
mesh(value,value,freq);
axis([-5 5 -5 5]);
zlabel('simulation');
title('Dependency check, inc (0,0.5) & inc(0.5,1.5)');
```

9.131

(a)

```
clear all;
close all;
s=100; %each second divided to 100 steps
%each sample is a gaussian random variable with zero mean, and variance
%alpha*t. Since alpha=1, then just generate randn(0,step)+prevvalue
n=3.5*s;
alpha=1;
maxrep=100;
x(1:maxrep,1:n)=0;
inc(1:maxrep,3)=0;
for r=1:1:maxrep
    x(r,1)=0;
    for i=2:1:n
        x(r,i)=x(r,i-1)+sqrt(1/s)*randn;
    end
    inc(r,1)=x(r,0.5*s)-x(r,1);
    inc(r,2)=x(r,1.5*s)-x(r,0.5*s+1);
    inc(r,3)=x(r,3.5*s)-x(r,1.5*s+1);
end

[y b]=hist(inc(:,1),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(0.5));
sno=sum(no);
figure(1);
plot(-6:0.4:6,no/sno,b,y/maxrep)
legend('normal pdf','simulation');

[y b]=hist(inc(:,2),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(1.5-0.5));
sno=sum(no);
figure(2);
plot(-6:0.4:6,no/sno,b,y/maxrep)
legend('normal pdf','simulation');

[y b]=hist(inc(:,3),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-8:0.4:8,0,sqrt(3.5-1.5));
sno=sum(no);
figure(3);
plot(-8:0.4:8,no/sno,b,y/maxrep)
legend('normal pdf','simulation');
```

(b)

```
clear all;
close all;
s=100; %each second divided to 100 steps
%each sample is a gaussian random variable with zero mean, and variance
%alpha*t. Since alpha=1, then just generate randn(0,step)+prevvalue
```

```
n=3.5*s;
alpha=1;
maxrep=100;
x(1:maxrep,1:n)=0;
inc(1:maxrep,3)=0;
for r=1:1:maxrep
    x(r,1)=0;
    for i=2:1:n
        x(r,i)=x(r,i-1)+sqrt(1/s)*randn;
    end
    inc(r,1)=x(r,0.5*s)-x(r,1);
    inc(r,2)=x(r,1.5*s)-x(r,0.5*s+1);
    inc(r,3)=x(r,3.5*s)-x(r,1.5*s+1);
end

[y b]=hist(inc(:,1),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(0.5));
sno=sum(no);
figure(1);
plot(-6:0.4:6,no/sno, '--',b,y/maxrep,'-o')
legend('normal','histogram');
title('inc(0,0.5)');
pdf1=no/sno;

[y b]=hist(inc(:,2),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(1.5-0.5));
sno=sum(no);
figure(2);
plot(-6:0.4:6,no/sno, '--',b,y/maxrep,'-o')
legend('normal','histogram');
title('inc(0.5,1.5)');
pdf2=no/sno;

[y b]=hist(inc(:,3),-6:0.4:6);
[sum(y/maxrep) mean(y/maxrep) var(y/maxrep)]
no=normal_pdf(-6:0.4:6,0,sqrt(3.5-1.5));
sno=sum(no);
pdf3=no/sno;
figure(3);
plot(-6:0.4:6,no/sno, '--',b,y/maxrep,'-o')
legend('normal','histogram');
title('inc(1.5,3.5)');

value=-6:0.4:6;
vallen=length(value);
%number of values is 12*2.5=30

chi=0;
p0(1:1:vallen,1:1:vallen)=0;
freq(1:1:vallen,1:1:vallen)=0;
for i=2:1:vallen-1
    for j=2:1:vallen-1
        p0(i,j)=pdf1(i)*pdf2(j);
        for r=1:1:maxrep
```

```
            if (inc(r,1) <= ((value(i)+value(i+1))/2) && inc(r,1) >
((value(i-1)+value(i))/2))
                if (inc(r,2) <= ((value(j)+value(j+1))/2) && inc(r,2) >
((value(j-1)+value(j))/2))
                    freq(i,j) = freq(i,j)+1;
                end
            end
        end
        freq(i,j)=freq(i,j)/maxrep;
        chi=chi+((freq(i,j)-p0(i,j))^2/p0(i,j));
    end
end
chi %(chi for 10000 sim is 0.02), (chi for 1000 sim is 0.35)
figure(4);
subplot(2,1,1);
mesh(value,value,p0);
axis([-5 5 -5 5]);
zlabel('theoretical prob');
subplot(2,1,2);
mesh(value,value,freq);
axis([-5 5 -5 5]);
zlabel('simulation');
title('Dependency check, inc (0,0.5) & inc(0.5,1.5)');
```

Problems Requiring Cumulative Knowledge

9.132

6.95 The increment of $X(t)$ in the interval $(t_1, t_2]$ has pdf:

$$f_{X(t_2)-X(t_1)}(x) = \frac{\lambda^{t_2-t_1}}{\Gamma(t_2-t_1)} x^{t_2-t_1-1} e^{-\lambda x}$$

a) We assume that $X(0) = 0$, then

$$\begin{aligned} f_{X(t_1)X(t_2)}(x,y) &= f_{X(t_1)}(x) f_{X(t_2)-X(t_1)}(y-x) \text{ by indep. increment property} \\ &= \frac{\lambda^{t_1}}{\Gamma(t_1)} x^{t_1} e^{-\lambda x} \frac{\lambda^{t_2-t_1}}{\Gamma(t_2-t_1)} (y-x)^{t_2-t_1-1} e^{-\lambda(y-x)} \\ &= \frac{\lambda^{t_2}}{\Gamma(t_1)\Gamma(t_2-t_1)} x^{t_1} (y-x)^{t_2-t_1-1} e^{-\lambda y} \end{aligned}$$

b)

$$\begin{aligned} R_X(t_1,t_2) &= E[X(t_1)X(t_2)] \qquad \text{assume } t_2 \geq t_1 \\ &= E[X(t_1)(X(t_2)-X(t_1)+X(t_1))] \\ &= E[X(t_1)^2] + E[X(t_1)]E[\underbrace{X(t_2)-X(t_1)}_{\text{increment}}] \end{aligned}$$

From Table 4.1

$$\begin{aligned} E[X(t_1)] &= \frac{\alpha}{\lambda} = \frac{t_1}{\lambda} \\ E[X^2(t_1)] &= VAR[X(t_1)] + E[X(t_1)]^2 \\ &= \frac{t_1}{\lambda^2} + \frac{t_1^2}{\lambda^2} \\ R_X(t_1,t_2) &= \frac{t_1}{\lambda^2} + \frac{t_1^2}{\lambda^2} + \frac{t_1}{\lambda}\left(\frac{t_2-t_1}{\lambda}\right) = \frac{t_1}{\lambda^2} + \frac{t_1 t_2}{\lambda^2} \\ &= \frac{t_1(1+t_2)}{\lambda^2} \qquad t_2 \geq t_1 \end{aligned}$$

If $t_1 \leq t_2$, then

$$R_X(t_1,t_2) = \frac{t_2(1+t_1)}{\lambda^2}$$

Note the similarities to the Wiener Process discussed in Ex. 6.38.

c) $R_X(t_1,t_2)$ is continuous at the point $t_2 = t_2 = t$ so $X(t)$ is M.S. continuous.

d)

$$R_X(t_1,t_2) = \begin{cases} \dfrac{t_2(1+t_1)}{\lambda^2} & t_1 < t_2 \\ \dfrac{t_2(1+t_2)}{\lambda^2} & g_2 \geq t_1 \end{cases}$$

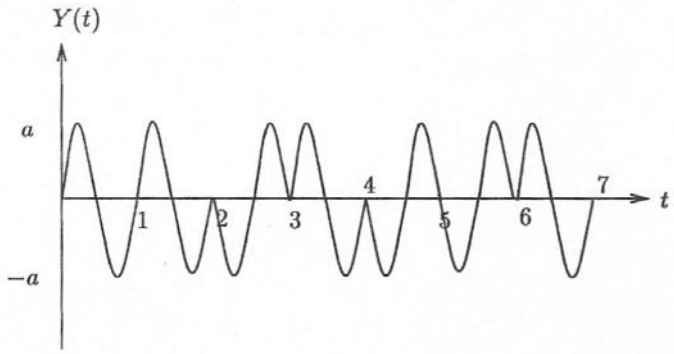
$$\frac{\partial R_X(t_1,t_2)}{\partial t_2} = \begin{cases} \dfrac{1+t_1}{\lambda^2} & t_1 \leq t_2 \quad \Rightarrow X(t) \text{ is } \underline{\text{NOT}} \\ \dfrac{t_1}{\lambda^2} & t_2 \geq t_1 \quad \text{M.S. differentiable} \end{cases}$$

$$= \frac{t_1}{\lambda^2} + \frac{1}{\lambda^2}u(t_1-t_2)$$

$$R_{X'}(t_1,t_2) = \frac{\partial^2 R_X(t_1,t_2)}{\partial t_1 \partial t_2} = \frac{1}{\lambda^2} + \frac{1}{\lambda^2}\delta(t_1-t_2)$$

This suggests that $X'(t)$ has this autocorrelation function if we generalize the notion of derivative of a random process.

9.133 a)



b) $f_{Y_1Y_2}(y_1, y_2)$

$$= \begin{cases} \frac{1}{2}\delta(y_1 - a\cos(2\pi t_1 + \frac{\pi}{2}))\delta(y_2 - a\cos(2\pi t_2 + \frac{\pi}{2})) \\ +\frac{1}{2}\delta(y_1 - a\cos(2\pi t_1 - \frac{\pi}{2}))\delta(y_2 - a\cos(2\pi t_2 - \frac{\pi}{2})), & nT \le t_1, t_2 < (n+1)T \\ \frac{1}{4}\delta(y_1 - a\cos(2\pi t_1 + \frac{\pi}{2}))\delta(y_2 - a\cos(2\pi t_2 + \frac{\pi}{2})) \\ +\frac{1}{4}\delta(y_1 - a\cos(2\pi t_1 - \frac{\pi}{2}))\delta(y_2 - a\cos(2\pi t_2 + \frac{\pi}{2})) \\ +\frac{1}{4}\delta(y_1 - a\cos(2\pi t_1 - \frac{\pi}{2}))\delta(y_2 - a\cos(2\pi t_2 + \frac{\pi}{2})) \\ +\frac{1}{4}\delta(y_1 - a\cos(2\pi t_1 - \frac{\pi}{2}))\delta(y_2 - a\cos(2\pi t_1 - \frac{\pi}{2})), & \begin{matrix} nT \le t_1 < (n+1)T \\ mT \le t_2 < (m+1)T, \quad m \ne n \end{matrix} \end{cases}$$

c)
$$\begin{aligned} E[Y(t)] &= \frac{1}{2}a\cos\left(2\pi t + \frac{\pi}{2}\right) + \frac{1}{2}a\cos\left(2\pi t - \frac{\pi}{2}\right) \\ &= -\frac{a}{2}\sin 2\pi t + \frac{a}{2}\sin 2\pi t \\ &= 0 \end{aligned}$$

$$\begin{aligned} &E[Y(t_1)Y(t_2)] \\ &= \frac{1}{2}a\cos\left(2\pi t_1 + \frac{\pi}{2}\right)a\cos\left(2\pi t_2 + \frac{\pi}{2}\right) + \frac{1}{2}a\cos\left(2\pi t_1 - \frac{\pi}{2}\right)\cos\left(2\pi t_1 - \frac{\pi}{2}\right) \\ &= \frac{a}{2}\sin(2\pi t_1)\sin(2\pi t_2) + \frac{a}{2}\sin(2\pi t_1)\sin(2\pi t_2) \\ &= a\sin(2\pi t_1)\sin(2\pi t_2), \qquad nT \le t_1, t_2 < (n+1)T \end{aligned}$$

$$E[Y(t_1)Y(t_2)] = E[Y(t_1)]E[Y(t_2)] = 0 \quad \text{otherwise}$$

d) $Y(t)$ is cyclostationary.

e) Yes.

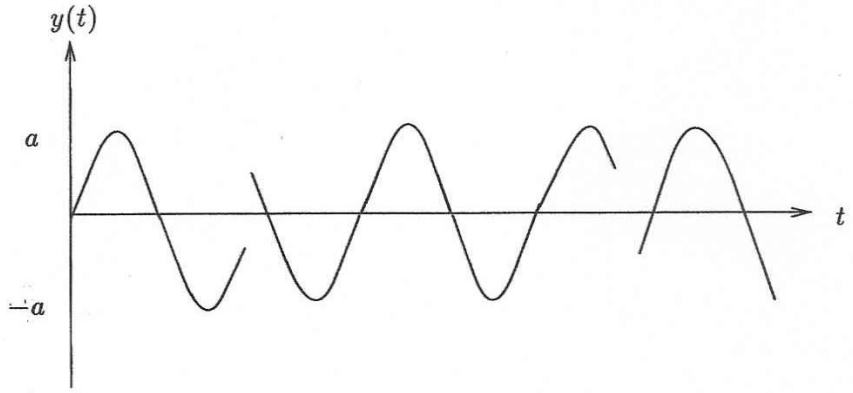
f) $X(t)$ is differentiable at all points $t \ne nT$.

$$E\left[\frac{dY(t)}{dt}\right] = \frac{d}{dt}E[Y(t)] = 0$$

$$\begin{aligned} R_{Y'}(t_1, t_2) &= \frac{\partial^2 R_Y(t_1, t_2)}{\partial t_1 \partial t_2} \\ &= \begin{cases} 4\pi^2 a\cos 2\pi t_1 \cos 2\pi t_2, & nT \le t_1, t_2 < (n+1)T \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

9.134

~~6.97~~ **a)** Each event in the Poisson process causes a sign reversal in $Y(t)$.



b) At time t_1, $Y(t)$ assumes the values $\cos 2\pi ft_1$ or $-\cos 2\pi ft_1$ depending on whether $N(t_1)$ is even or odd. Similarly $Y(t_2)$ is $\cos 2\pi ft_2$ or $-\cos 2\pi ft_2$ depending on whether an even or odd number of events occur in the interval $(t_1, t_2]$. Thus from Example 6.22 we know that

$$\begin{aligned} P[\text{even \# events in } t \text{ seconds}] &= \frac{1}{2}\{1 + e^{-2\alpha t}\} \triangleq \pi_e(t) \\ P[\text{odd \# events in } t \text{ seconds}] &= \frac{1}{2}\{1 - e^{-2\alpha t}\} \triangleq \pi_o(t) \end{aligned}$$

and

$$\begin{aligned} f_{Y(t_1)Y(t_2)}(y_1, y_2) = {} & \pi_e(t_1)\delta(y_1 - \cos 2\pi ft_1)[\pi_e(t_2 - t_1)\delta(y_2 - \cos 2\pi ft_2) \\ & + \pi_o(t_2 - t_1)\delta(y_2 + \cos 2\pi ft_2)] \\ & + \pi_o(t_1)\delta(y_1 + \cos 2\pi ft_1)[\pi_e(t_2 - t_1)\delta(y_2 + \cos 2\pi ft_2)] \\ & + \pi_o(t_2 - t_1)\delta(y_2 - \cos 2\pi ft_2)\ . \end{aligned}$$

c)
$$\begin{aligned} E[Y(t)] &= \cos 2\pi ft\pi_e(t) - \cos 2\pi ft\pi_o(t) \\ &= (\cos 2\pi ft)e^{-2\alpha t} \\ E[Y(t_1)Y(t_2)] &= \Sigma\Sigma y_1 y_2 P[y_1, y_2] \\ &= \cos 2\pi ft_1 \cos 2\pi ft_2 \pi_e(t_1)\pi_e(t_2 - t_1) \\ &\quad - \cos 2\pi ft_1 \cos 2\pi ft_2 \pi_e(t_1)\pi_o(t_2 - t_1) \\ &\quad - \cos 2\pi ft_1 \cos 2\pi ft_2 \pi_o(t_1)\pi_o(t_2 - t_1) \\ &\quad + \cos 2\pi ft_1 \cos 2\pi ft_2 \pi_o(t_1)\pi_e(t_2 - t_1) \end{aligned}$$

$$\begin{aligned} &= \cos 2\pi f t_1 \cos 2\pi f t_2 (\pi_e(t_2 - t_1) - \pi_o(t_2 - t_1)) \\ &= \cos 2\pi f t_1 \cos 2\pi f t_2 e^{-2\alpha(t_2 - t_1)} \\ &= \frac{1}{2}\cos 2\pi f(t_2 - t_1)e^{-2\alpha(t_2-t_1)} + \frac{1}{2}\cos 2\pi f(t_2 + t_1)e^{-2\alpha(t_2-t_1)} \end{aligned}$$

If $t_1 \geq t_2$ we have

$$E[Y(t_1)Y(t_2)] = \cos 2\pi f t_1 \cos 2\pi f t_2 e^{-2\alpha(t_1 - t_2)}.$$

Thus

$$E[Y(t_1)Y(t_2)] = \cos 2\pi f t_1 \cos 2\pi f t_2 e^{-2\alpha|t_2 - t_1|}.$$

d) $E[Y(t)]$ varies with time so $Y(t)$ is not stationary. $Y(t)$ does not depend solely on $t_2 - t_1$ so it is not WSS. If we let $t_2 = t_1 + \tau$ and let $t_1 \to \infty$, $E[Y(t_1)Y(t_2)]$ still does not depend solely on $t_2 - t_1$ so it is not asymptotically WSS.

If we consider $t_1 + mT$ and $t_2 + mT$ we have

$$\begin{aligned} E[Y(t_1 + mT)Y(t_2 + mT)] &= \cos 2\pi f(t_1 + mT)\cos 2\pi f(t_2 + mT)e^{-2\alpha|t_2 - t_1|} \\ &= \cos 2\pi f t_1 \cos 2\pi f t_2 e^{-2\alpha|t_2 - t_1|} \\ &= E[Y(t_1)Y(t_2)] \end{aligned}$$

However the mean is:

$$\begin{aligned} E[Y(t_1 + mT)] &= \cos 2\pi f(t_1 + mT)e^{-2\alpha(t_1 + mT)} \\ &\neq E[Y(t_1)]\,. \end{aligned}$$

As $t_1 \to \infty$ both of these mean terms approach zero. We conclude that $Y(t)$ is asymptotically wide sense cyclostationary.

e) $R_Y(t_1, t_2)$ is continuous in t_1 and t_2 so $Y(t)$ is M.S. continuous.

f)
$$R_Y(t_1, t_2) = \begin{cases} \cos 2\pi f t_1 \ \cos 2\pi f t_2 e^{-2\alpha(t_2 - t_1)} & t_2 \geq t_1 \\ \cos 2\pi f t_1 \ \cos 2\pi f t_2 e^{2\alpha(t_2 - t_1)} & t_1 < t_2 \end{cases}$$

We see that $R_Y(t_1, t_2)$ has a cusp at $t_1 = t_2$ so $Y(t)$ is not M.S. differentiable.

9.135

6.98 Assume $h(t) = 0$ for $t < 0$.

a) We condition on the number of occurrence, $N(t)$, up to t

$$\begin{aligned}
E[X(t)] &= E[E[X(t)|N(t)]] \\
E[X(t)|N(t)=k] &= E\left[\sum_{j=1}^{k} A_j h(t-s_j)\right] \\
&= \sum_{j=1}^{k} E[A_j]E[h(t-s_j)] \\
E[h(t-s_j)] &= \int_0^t h(t-s)\cdot\frac{1}{t}ds \\
&= \int_0^t h(u)\frac{1}{t}du \\
E[X(t)|N(t)=k] &= E[A_j]\cdot k\cdot\int_0^t h(u)\frac{1}{t}du \\
E[X(t)] &= E[E[X(t)|N(t)]] \\
&= E\left[E[A_j]\frac{N(t)}{t}\int_0^t h(u)du\right] \\
&= E[A_j]\lambda\int_0^t h(u)du
\end{aligned}$$

For the autocorrelation function, we condition on the number of occurrences at t_1 and t_2

$$E[X(t_1)X(t_2)] = E[E[X(t_1)X(t_2)|N(t_1)=k,\ N(t_2)=n]],\quad t_1 \le t_2$$

$$\begin{aligned}
E[X(t_1)X(t_2)|N(t_1)=k, N(t_2)=k+n] &= E\left\{\sum_{j=1}^{k} A_j h(t_1-s_j)\sum_{l=1}^{k+n} A_l h(t_2-s_l)\right\} \\
&= \sum_{j=1}^{k}\sum_{l=1}^{k+n} E[A_jA_l]E[h(t_1-s_j)h(t_2-s_l)] \\
E[A_jA_l] &= \begin{cases} E[A_j^2] & j=l \\ E[A_j]E[A_l] & j\neq l \end{cases} \\
E[h(t_1-s_j)h(t_2-s_l)] &= \begin{cases} E[h(t_1-s_j)h(t_2-s_j)] & j=l \\ E[h(t_1-s_j)]E[h(t_2-s_l)] & j\neq l \end{cases}
\end{aligned}$$

When $j = l$, s_j occurs in $(0, t_1)$:

$$E[h(t_1-s_j)h(t_2-s_j)] = \int_0^{t_1} h(t_1-s)h(t_2-s)\frac{ds}{t_1}$$

Substitution into first equation gives:

$$\begin{aligned}E[X(t_1)X(t_2)|N(t_1)=k, N(t_2)=k+n] &= kE[A^2]\int_0^{t_1} h(t_1-s)h(t_2-s)\frac{ds}{t_1} \\ &\quad +k(k-1)E[A]^2\int_0^{t_1} h(t_1-s)\frac{ds}{t_1}\int_0^{t_1} h(t_2-s)\frac{ds}{t_1} \\ &\quad +k\cdot nE[A]^2\int_0^{t_1} h(t_2-s)\frac{ds}{t_1}\int_{t_1}^{t_2} h(t_2-s)\frac{ds}{t_2-t_1}\end{aligned}$$

Now

$$\begin{aligned}E[N(t_1)] &= \lambda t_1 \qquad E[N(t_1)(N(t_1)-1] = \lambda^2 t_1^2 \\ E[N(t_2)-N(t_1)] &= \lambda(t_2-t_1)\end{aligned}$$

Thus

$$\begin{aligned}E[X(t_1)X(t_2)] &= \lambda\int_0^{t_1} h(t_1-s)h(t_2-s)ds \\ &\quad +\lambda^2E[A]^2\int_0^{t_1} h(t_1-s)ds\int_0^{t_1} h(t_2-s)ds \\ &\quad +\lambda^2E[A]^2\int_0^{t_1} h(t_2-s)ds\int_{t_1}^{t_2} h(t_2-s)ds \\ &= \lambda\int_0^{t_1} h(t_1-s)h(t_2-s)ds \\ &\quad +\lambda^2E[A]^2\int_0^{t_1} h(t_1-s)ds\int_0^{t_2} h(t_2-s)ds\end{aligned}$$

b) $h(t)=u(t)$

$$E[X(t)] = E[A]\lambda\int_0^t h(u)du - E[A]\lambda t$$

This is consistent with the expected value of Poisson RV when $A=1$

$$\begin{aligned}E[X(t_1)X(t_2)] &= \lambda t_1E[A^2]\frac{1}{t_1}\int_0^{t_1} |\cdot|du \\ &\quad +\lambda^2t_1^2E[A]^2\frac{1}{t_1^2}\int_0^{t_1} du\int_0^{t_1} du \\ &\quad +\lambda(t_2-t_1)\lambda t_1E[A^2]\frac{1}{t_1}\int_0^{t_1} du\frac{1}{t_2-t_1}\int_0^{t_2-t_1} du \\ &= \lambda t_1E[A^2]+\lambda^2t_1^2E[A^2]+\lambda(t_2-t_1)\lambda t_1E[A^2] \\ &= \lambda t_1E[A^2]+(\lambda^2t_1t_2)E[A^2] \qquad (t_1\le t_2)\end{aligned}$$

This is consistent again with the autocorrelation function of Poisson RV when $A=1$.

c) $h(t)=p(t)$, a rectangular pulse of duration T

$$\begin{aligned}E[X(t)] &= E[A]\lambda\int_0^t p(u)du \\ &= \begin{cases} E[A]\lambda T & \text{if } t\ge T \\ E[A]\lambda t & \text{if } t<T\end{cases} \\ &= E[A]\min(t,T) \\ E[X(t_1)X(t_2)] &= \lambda t_1E[A^2]\frac{1}{t_1}\int_0^{t_1} p(u)h(t_2-t_1+u)du \\ &\quad +\lambda^2t_1^2E[A]^2+\frac{1}{t_1^2}\min(t_1,T)\int_0^{t_1} h(t_2-t_1+u)du \\ &\quad +\lambda(t_2-t_1)\lambda t_1E[A]^2\frac{1}{t_1}\min(t_1,T)\frac{1}{t_2-t_1}\min(t_2-t_1,T) \\ &\qquad \text{for } t_1\le t_2\end{aligned}$$

9.136

6.99 a)

$$\begin{aligned}
E[X(t)] &= E[A_1\cos(\omega_0 t+\theta_1)] + E[A_2\cos(\sqrt{2}\omega_2 t+\theta_2)] \\
&= E[A_1]E[\cos(\omega_0 t+\theta_1)] + E[A_2]E[\cos(\sqrt{2}\omega_0 t+\theta_2)] \\
&= 0 \\
E[X(t_1)X(t_2)] &= E[\{A_1\cos(\omega_0 t_1+\theta_1) + A_2\cos(\omega_0 t_1+\theta_2)\}]\cdot \\
&\quad \{A_1\cos(\sqrt{2}\omega_0 t_2+\theta_1) + A_2\cos(\sqrt{2}\omega_0 t_2+\theta_2)\} \\
&= E[A_1^2\cos(\omega_0 t_1+\theta_1)\cos(\sqrt{2}\omega_0 t_2+\theta_1)] \\
&\quad +E[A_2A_1\cos(\omega_0 t_1+\theta_2)\cos(\sqrt{2}\omega_0 t_2+\theta_1)] \\
&\quad +E[A_1A_2\cos(\omega_0 t_1+\theta_1)\cos(\sqrt{2}\omega_0 t_2+\theta_2)] \\
&\quad +E[A_2^2\cos(\omega_0 t_1+\theta_2)\cos(\sqrt{2}\omega_0 t_2+\theta_2)] \\
&= \frac{1}{2}E[A_1^2]\cos(\omega_0 t_1-\sqrt{2}\omega_0 t_2) + \frac{1}{2}E[A_2^2]\cos(\omega_0 t_1-\sqrt{2}\omega_0 t_2) \\
&= \frac{1}{2}\{E[A_1^2]+E[A_2^2]\}\cos\omega_0(t_1-\sqrt{2}t_2)
\end{aligned}$$

b) If $X(t)$ were mean-square periodic then its autocorrelation would depend only on $\tau = t_2 - t_1$ and it would be periodic in τ. The $E[X(t_1)X(t_2)]$ above does not satisfy these properties, so $X(t)$ is not mean square periodic.

c) If we condition on Θ_1 and Θ_2 then $X(t_1)$ and $X(t_2)$ are defined by a linear transformation on A_1 and A_2:

$$\begin{bmatrix} X(t_1) \\ X(t_2) \end{bmatrix} = \begin{bmatrix} \cos(\omega_0 t_1+\theta_1) & \cos(\sqrt{2}\omega_0 t_1+\theta_2) \\ \cos(\omega_0 t_2+\theta_1) & \cos(\sqrt{2}\omega_0 t_2+\theta_2) \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \Gamma(t_1,t_2,\theta_1,\theta_2)\underline{A}$$

Since A_1 and A_2 are jointly Gaussian, the conditional joint pdf of $X(t_1)$ and $X(t_2)$ are also jointly Gaussian with correlation matrix

$$K_X(t_1,t_2,\theta_1,\theta_2) = \Gamma(t_1,t_2,\theta_1,\theta_2)K_A\Gamma^*(t_1,t_2,\theta_1,\theta_2)$$

where K_A is the covariance matrix of $\underline{A}$. The joint pdf of $X(t_1)$ and $X(t_2)$ is then found by averaging over the random phases Θ_1 and Θ_2:

$$f_{X(t_1)X(t_2)}(x_1,x_2) = \int_0^{2\pi}\int_0^{2\pi} \frac{e^{-\frac{1}{2}\underline{x}^+(\Gamma K_A\Gamma^+)^{-1}\underline{x}}}{2\pi|\Gamma K_A\Gamma^+|^{1/2}}\frac{d\theta_1}{2\pi}\frac{d\theta_2}{2\pi}$$

where we assumed that $\underline{A}$ (and hence $\underline{X}$) is zero mean. A slightly simpler expression is obtained if we write the joint characteristic function

$$\Phi_{X(t_1)X(t_2)}(\omega_1,\omega_2) = \int_0^{2\pi}\int_0^{2\pi} e^{-\frac{1}{2}\underline{\omega}^+\Gamma K_A\Gamma^+\underline{\omega}}\frac{d\theta_1}{2\pi}\frac{d\theta_2}{2\pi}.$$

9.137

6.100 a) Since $X(t)$ is Markov for $X(t_1) = x_1$, $X(t_2) = x_2$, $X(t_3) = x_3$, we have

$$f(x_3, x_1|x_2)\frac{f(x_1, x_2)f(x_3|x_2, x_1)}{f(x_2)} = f(x_3|x_2)f(x_1|x_3)$$

$$\begin{aligned}
C_X(t_3, t_1) &= E[X_3X_1] - m_3m_1 \\
&= E[E[X_3X_1|X_2]] - m_3m_1 \\
&= E[E[X_3|X_2]E[X_1|X_2]] - m_3m_1 \\
&= E\left[\left\{m_3 + \rho_{2,3}\frac{\sigma_3}{\sigma_2}(X_2 - m_2)\right\}\left\{m_1 + \rho_{1,2}\frac{\sigma_1}{\sigma_2}(X_2 - m_2)\right\}\right] - m_3m_1 \\
&= E\left[\rho_{2,3}\frac{\sigma_3}{\sigma_2}(X_2 - m_2)\rho_{1,2}\frac{\sigma_1}{\sigma_2}(X_2 - m_2)\right] \\
&= \frac{\rho_{2,3}\sigma_3\sigma_2\rho_{1,2}\sigma_1\sigma_2}{\sigma_2\sigma_2} \\
&= \frac{C_X(t_3, t_2)C_X(t_2, t_1)}{C_X(t_2, t_2)} \qquad (t_1 \le t_2 \le t_3)
\end{aligned}$$

b) Wiener Process

$$\begin{aligned}
C_X(t_3, t_2) &= \alpha t_2, \qquad C_X(t_2, t_1) = \alpha t_1, \qquad C_X(t_3, t_1) = \alpha t_1 \\
C_X(t_3, t_2) &= \frac{C_X(t_3, t_2)C_X(t_2, t_1)}{C_X(t_2, t_2)}
\end{aligned}$$

So Wiener process is Gauss-Markov.

For Ornstein-Uhlenbeck process

$$\begin{aligned}
\frac{C_X(t_3, t_2)C_X(t_2, t_1)}{C_X(t_2, t_2)} &= \frac{\sigma^2}{2\alpha}\frac{(e^{-\alpha(t_3-t_2)} - e^{-\alpha(t_3+t_2)})(e^{-\alpha(t_2-t_1)} - e^{-\alpha(t_2+t_1)})}{e^{-\alpha(t_2-t_2)} - e^{-\alpha(t_2+t_2)}} \\
&= \frac{\sigma^2}{2\alpha}\frac{e^{-\alpha(t_3-t_1)} - e^{-\alpha(t_3+2t_2-t_1)} - e^{-\alpha(t_3+t_1)} + e^{-\alpha(t_3+2t_2+t_1)}}{1 - e^{-2\alpha t_2}} \\
&= \frac{\sigma^2}{2\alpha}(e^{-\alpha(t_3-t_1)} - e^{-\alpha(t_3+t_1)})
\end{aligned}$$

So Ornstein-Uhlenbeck process is also Gauss-Markov.

9.138

6.101 a) $Y_{4n+1} = A_{2n+1}$, $Y_{4n+2} = A_{2n+2}$, $Y_{4n+1} = B_{2n+1}$, $Y_{4n+4} = B_{2n+2}$, $n = 0, 1, \ldots$

$$\begin{aligned} E[Y_{4j+k}Y_{4m+n}] &= E[A_{2j+k}A_{2m+n}] = \sigma_1^2 \rho_1^{|2m+n-2j-k|} \quad 1 \le k,\ n \le 2 \\ E[Y_{4j+k}Y_{4m+n}] &= E[B_{2j+k}B_{2m+n-2}] = \sigma_1^2 \rho_1^{|2m+n-2j-k|} \quad 3 \le k,\ n \le 4 \\ E[Y_{4j+k}Y_{4m+n}] &= 0 \quad \text{otherwise} \end{aligned}$$

b) Y_m is not WS stationary, but is cyclostationary.

c) $m = 4n + 1$

$$f_{Y_m Y_{m+1}}(y_m, y_{m+1}) = f_{A_{2n+1}A_{2n+2}}(y_m, y_{m+1}) \sim N(0, 0, \sigma_1^2, \sigma_1^2, \rho_1\sigma_1^2)$$

$m = 4n + 3$

$$f_{Y_m Y_{m+1}}(y_m, y_{m+1}) = f_{B_{2n+1}B_{2n+2}}(y_m, y_{m+1}) \sim N(0, 0, \sigma_2^2, \sigma_2^2, \rho_2\sigma_2^2)$$

$m = 4n + 2$

$$f_{Y_m Y_{m+1}}(y_m, y_{m+1}) = f_{A_{2n+2}B_{2n+1}}(y_m, y_{m+1}) \sim N(0, 0, \sigma_1^2, \sigma_2^2, 0)$$

$m = 4n + 4$

$$f_{Y_m Y_{m+1}}(y_m, y_{m+1}) = f_{B_{2n+2}A_{2n+3}}(y_m, y_{m+1}) \sim N(0, 0, \sigma_2^2, \sigma_1^2, 0)$$

d) $Z_m = Y_{m+T}$, will "stationarize" Y_m.

$$\begin{aligned} E[Z_m Z_n] &= E[Y_{m+T}T_{n+T}] \\ &= \sum_{T=0}^{3} E[Y_{m+T}Y_{n+T}] \cdot \frac{1}{4} \end{aligned}$$

Z_m is stationary.

$$f_{Z_m Z_{m+1}}(z_m, z_{m+1}) = \sum_{T=0}^{3} f_{Y_{m+T}Y_{m+1+T}}(z_m, z_{m+1}) \cdot \frac{1}{4}$$

9.139

.102 $V_m = A_{2m-1}$

a) $E[V_m V_n] = E[A_{2m-1}A_{2n-1}] = \sigma_1^2 \rho_1^{2|m-n|}$

b) $f_{V_m,V_{m+k}}(v_m, v_{m+k}) \sim N(0, 0, \sigma_1^2, \sigma_1^2, \sigma_1^2 \rho_1^{2|k|})$

c) $W_{2n} = 0$, $W_{2n+1} = A_{2n+1}$
$E[W_m W_n] = 0$ if m is even or n is even.
$E[W_m W_n] = \sigma_1^2 \rho_1^{|m-n|}$ if both m and n are odd.
W_n is not a Gaussian process.

9.140

.103

$$Y_{2n} = \frac{1}{\sqrt{2}}A_{2n} + \frac{1}{\sqrt{2}}A_{2n+1}$$
$$Y_{2n+1} = \frac{1}{\sqrt{2}}A_{2n} - \frac{1}{\sqrt{2}}A_{2n+1}$$

a)
$$E[Y_{2m}Y_{2n}] = \frac{1}{2}E[(A_{2m} + A_{2m+1})(A_{2n} + A_{2n+1})] = 0, \quad \text{if } m \neq n$$
$$E[Y_{2n}^2] = \frac{1}{2}E[(A_{2n} + A_{2n+1})^2] = 1$$
$$E[Y_{2m+1}Y_{2n+1}] = 0 \quad \text{if } m \neq n$$
$$E[Y_{2n+1}^2] = 1$$
$$E[Y_{2m}Y_{2n+1}] = 0 \quad \text{if } m \neq n$$
$$E[Y_{2n}Y_{2n+1}] = \frac{1}{2}E[A_{2n}^2 - A_{2n+1}^2] = 0$$
so $$E[Y_m Y_n] = \delta m, n$$

b) Y_n is a stationary random process.

c) Y_n, Y_{n+1}, Y_{n+2} are independent Gaussian, random variables.

9.141

9.104 $N = X_1 + X_2 + ... + X_n$

$$P[X_i = 1] = p, \quad P[X_i = 0] = 1 - p$$

$$S_n = Y_1 + Y_2 + ... + Y_n$$

Y_i is exponential if $X_i = 1$, or with probability p.
Y_i is zero if $X_i = 0$, or with probability $1 - p$.

a)

$$\begin{aligned} E[Y] &= E[E[Y|X]] \\ E[Y|X=1] &= \frac{1}{\lambda}, \quad E[Y|X=0] = 0 \\ E[Y] &= p \cdot \frac{1}{\lambda} + (1-p)0 = p \cdot \frac{1}{\lambda} \end{aligned}$$

Similarly

$$\begin{aligned} E[Y^2] &= p \cdot \left(\frac{1}{\lambda^2} + \frac{1}{\lambda^2}\right) = p \cdot \frac{2}{\lambda^2} \\ E[S_n] &= nE[Y] = np\frac{1}{\lambda} \end{aligned}$$

Assume $m \leq n$

$$\begin{aligned} E[S_m S_n] &= E[(Y_1 + ... + Y_m)(Y_1 + ... + Y_m + ... + Y_n)] \\ &= mE[Y^2] + m(m-1)E[Y_iY_j] + (n-m) \cdot mE[Y_kY_l] \quad (i \neq j, \quad k \neq l) \\ &= m \cdot p\frac{1}{\lambda^2} + m(m-1) \cdot \left(p\frac{1}{\lambda^2}\right)^2 + (n-m)m\left(p\frac{1}{\lambda}\right)^2 \\ &= mp\frac{1}{\lambda^2} + m(n-1)p^2\frac{1}{\lambda^2} \end{aligned}$$

b) No, the mean square linearly with n.

c) Yes, the process has independent increments and is therefore Markov.

d) $f_{S_n S_{n+m}}(x, y) = f_{S_n}(x) f_{S_m}(y - x)$ where

$$\begin{aligned} f_{S_n}(x) &= \sum_{k=0}^{n} f_{S_n}(x|N=k)P[N=k] \\ &= \sum_{k=0}^{n} \frac{\lambda e^{-\lambda x}(\lambda x)^{k-1}}{(k-1)!} \binom{n}{k} p^k (1-p)^{n-k} \end{aligned}$$